

In assembling this unit, the bevel driver should be laid in the housing with the gear with the widest face at the cupped or recessed end of the housing. The two halves of the housing must be properly put together also.

TO ASSEMBLE THE ENGINE

CRANK CASE, UPPER HALF

1. Place the upper half of the crank case in the inspecting stand bottom side up.
2. Wipe out all bearings carefully.

CRANK SHAFT

3. Oil all bushings with fresh, clean oil and lay the crank shaft with its thrust bearing in place. Be sure that the thrust bearing is properly seated in the case and that the sleeves do not permit end play.

NOTE.—The thrust bearing sleeves are made in different thicknesses, No. 12252 being the thick one and No. 12238 the thin one. A combination of these sleeves may be obtained, which will insure the proper fit in the case.

CONNECTING RODS

4. The connecting rods may now be assembled on the crank shaft. The rods are numbered on the web at the piston end, 1, 2, 3, etc., indicating their location on the crank shaft beginning at the gear end. The side of the rods on which the numbers are stamped should be toward the gear end of the shaft.

All forked rods should be so fitted that they will be on the right-hand side of the engine looking from the gear end toward the propeller end when the engine is right side up. Crank pins should be liberally oiled before the rods are put in place, as should also the outside of the connecting rod bushings before the plain end rods are assembled. The hollow crank pins and main bearings should be filled with clean oil.

Draw up all connecting rod bolts snugly and cotter-pin them.

NOTE.—The cotter pins in the bolts through the forked end rod should be inserted so that the heads will be on the inside or toward the plain end rod. Otherwise the bent-over ends of the cotter pins will interfere with the plain end rod.

Be sure that none of these bolts are overlooked.

CRANK CASE, LOWER HALF

The lower half of the crank case may now be put in place.

5. Put the washers No. 115 and nuts No. 105 on the front and rear bearing bolts and draw them tight.

WATER-PUMP BEVEL DRIVER

6. Put the water-pump bevel driver with its bushing in place and fasten with the lock screw No. 174. Test the backlash of the water-pump bevel driver gear and the crank-shaft gear.

This should be 0.005 inch minimum and 0.010 inch maximum.

If these gears mesh too tightly, shims No. 8502, 0.002 inch thick, may be placed between the crank-shaft gear and the crank-shaft flange in sufficient number to produce the proper backlash.

7. Now turn the whole assembly right side up and complete the installation of the bearing-bolt washers No. 115 and nuts No. 105. Draw up these nuts snugly and cotter-pin them.

8. Install bolts No. 116 and tighten and cotter-pin them.

PISTONS

9. Proceed to fit the pistons on their proper rods. Remember that the pistons are marked right and left and numbered 1, 2, 3, 4, 5, and 6, and that the pistons should be so placed on the rods that the numbers are toward the gear end of the engine.

The piston pins should be oiled before being pressed into place.

10. Insert the piston-pin retainers No. 12547. These should be a light tapping fit in the piston bosses.

CYLINDERS

11. Oil the inside walls of the cylinders and the outside of the pistons. The cylinders are numbered 1, 2, 3, 4, etc., and are marked right and left, indicating their position on the crank case. These marks will be found on the edge of the base flange near the water inlet. Be sure that the gasket No. 12346 is in place on the crank-case studs. The crank should be turned over so that the piston on which the cylinder is being placed is at the top of its stroke.

The rings should be compressed in the grooves as the cylinder is slipped down over them.

12. Push the cylinder carefully down into place, put on the nuts No. 103 and run them down on the studs, but do not tighten them.

13. After all the cylinders are in place put on the gaskets No. 8175 and the intake headers No. 12045.

Caution.—Insert loose-fitting wooden plugs in spark-plug holes to prevent any small parts from falling into cylinders.

INTAKE HEADERS

14. Now put on the intake-head stud washers No. 113 and the intake-header nuts No. 103. The intake-header nuts and the cylinder to crank-case nuts on each set of three cylinders should all be tightened up at the same time; i. e., each nut should be drawn down one after another about one-fourth turn at a time until all are tight with an equal tension.

15. The special nuts No. 8234 may now be put in place and tightened up with a special long-shanked socket wrench.

16. Tighten all nuts and properly cotter-pin them.

17. Put the gaskets No. 8172 between the intake header and the water-inlet elbows No. 8173 and tighten these up with the cap screws No. 201. The gaskets No. 8172 should be soaked in water before putting them in place. Leave three screws No. 201 on each side (one at each end of the engine and one about midway between these two) loose until the cable tube has been installed.

OUTLET HEADERS

18. Put the water-outlet header gaskets No. 8153 in place and bolt on the front and rear water outlets No. 12402 and 12401, respectively. Before these water outlets are bolted down the extension tube No. 12363 and the two pieces of hose No. 12128 should be slipped in place.

19. Now fasten down the water outlets with nuts No. 101 and cotter-pin them.

20. Slip the carburetor to intake-header bolts No. 8398–No. 8399 in place.

CARBURETORS

21. Attach the two carburetors. Be sure that the unions No. 12367 and gaskets No. 182 are in place. With respect to these gaskets it is advisable to fit new ones after an engine has been overhauled.

Care should be taken that the carburetors are mounted in their proper position on the engine (propeller end and gear end) and that the air intake scoops open toward the propeller end.

THROTTLES

22. The carburetor throttle coupling shaft No. 12498 may now be put in place and the two sets of throttles synchronized as follows:

Screw out the throttle stop screw on each carburetor until it clears the gear sector when the throttles are entirely closed. Slack off the adjusting screws No. 13416 until the lever No. 12498 clears each one of them by approximately $\frac{1}{16}$ inch.

Now entirely close the throttles on both carburetors and holding them firmly in a closed position screw up the adjusting screws No. 13416 until they just touch each side of the lever No. 12498 and lock the screws by means of a wire through the head of each. Connect up the altitude adjustment coupling rod in such a manner that both altitude adjustments will get the full throw in each direction.

TIMING ENGINE

A part of the equipment of every Liberty engine is a timing disk, so designed that it may be mounted on the flywheel as a permanent

fixture or used only in the hangar or repair shop as a means of checking the setting of valves and distributors. To use this timing disk, proceed as follows:

1. If the timing disk is not already mounted on the flywheel install it in such a manner that the dowel in the propeller-hub flange enters the dowel hole in the disk. It may be clamped in this position by means of two bolts through the propeller-hub bolt holes.

2. Remove the spark plug from the flywheel of No. 6 L cylinder.

3. Insert a pencil or scale through the spark-plug hole and turn the engine over until the piston on its upstroke touches the pencil and causes it to ride up. Continue to turn the engine over slowly until the piston as indicated by the travel of the pencil stops moving upward and is just about to start down. This will be approximately the top dead center.

4. Allow the crank shaft to remain in this position temporarily and clamp the timing pointers, which will be found in the tool kit under the special cylinder base flange nuts No. 8234, so that the pointers extend over the edge of the timing disk.

5. With the end of the pencil resting on the top of the piston make a mark with a knife blade about one-half inch above the edge of the spark-plug hole.

6. Turn the engine over in a forward direction until the pencil has moved down so that the mark is even with the top edge of the hole, and with a piece of chalk or a pencil mark the disk in line with one of the pointers.

7. Turn the engine backward until the pencil has moved up and down to the point where the mark is again even with the top of the spark-plug hole, and mark the disk in line with the pointer.

8. With a pair of dividers find the point midway between the two marks on the disk. This point will indicate the exact dead center of No. 1 and No. 6 cranks and should be marked with chalk or pencil.

9. Turn the engine over until this dead-center mark is in line with the pointer. Allow the crank shaft to remain in this position; and

10. Reset the pointers so that they come in line with the dead-center marks stamped on the disk.

11. Turn the engine over in the direction of rotation through 10° as indicated by the scale on the disk.

NEUTRAL POINT

The crank shaft is now set on the neutral point of No. 6 left cylinder and the firing point-spark retarded of No. 1 left cylinder. Reference to the timing instructions will explain what is meant by "neutral point." It is the point 10° past the top dead center which marks the beginning and end of the cycle of operations. The exhaust valve closes and the inlet valve opens at this point.

Mount the generator drive-shaft assembly No. 8210, being careful that the gasket No. 8347 is in place and a sufficient number of shims No. 8511 (0.002 inch thick) to insure proper mesh of the generator drive-shaft lower gear with the crank-shaft gear. These gears should have a minimum back lash of 0.005 inch and a maximum of 0.010 inch.

Mount the two cam-shaft drive shafts No. 8208, meshing the gears in such a manner that the mark on the splined coupling is "fore and aft" or parallel with the center line of the engine.

Now mount the cam-shaft housing assemblies.

If it was not found necessary to replace either the cam shaft or gear, be sure that the marked teeth on both gear and pinion are in line. This should bring the mark on the splined end of the drive shaft "fore and aft."

The assemblies may now be set in place with the splined coupling marks in line.

NOTE.—All marks for both right and left cylinders are located with No. 1–6 cranks 10° past left dead center.

Complete the installation of these assemblies by replacing the washers No. 112 and the nuts No. 102 and properly cotter pinning them.

Slip the felt washers No. 8068 into place and tighten up the stuffing boxes No. 8066.

TAPPET GAP AND FIRING POINT

Test the gap between all tappets and the valves which they operate. The tappet gap for each cylinder should be checked when that cylinder is on the firing point. The firing point of No. 1 cylinder is the neutral point of No. 6 on the same side. The firing point of No. 2 is the neutral point of No. 5. The firing point of No. 3 is the neutral point of No. 4. The firing point of No. 4 is the neutral point of No. 3. The firing point of No. 5 is the neutral point of No. 2. The firing point of No. 6 is the neutral point of No. 1. It will be noticed that the sum of the numbers of these pair of cylinders is always seven. For example: to find the firing point of No. 4 cylinder, turn the engine over, meanwhile watching the No. 3 exhaust valve. When this valve has just closed and before No. 3 inlet valve has opened, the neutral point of No. 3 cylinder will have been reached. This will be the firing point of No. 4. With the engine cold the clearance between the inlet valve tappets and the valve stems should be 0.014 to 0.016. The clearance between the exhaust valve tappets and valve stems should be 0.019 to 0.021. This clearance should be adjusted by adding or taking out shims under the tappet head. These shims are made in varying thicknesses; the thick shim No. 8089 being 0.015 inch thick, the medium shim No. 8086 being 0.008 inch, and the thin shim No. 8087 being 0.003 inch. The combination of

these shims will permit of a very accurate adjustment of the gap. Be sure that the shims are properly placed and that the nuts on the tappets are tightly drawn up and cotted.

If it was found necessary to replace either the cam shaft or the cam-shaft gear, the gear should have been left unmounted as instructed. In this case proceed as follows:

1. With No. 1-6 crank set 10° past the left dead center, the marked splines on the cam-shaft drive shaft set "fore and aft," the marked splines on the upper cam-shaft drive shaft in line with them, the marked tooth on the upper cam-shaft drive shaft gear should be toward the observer and on the center line of the cylinders.

2. Without moving any of this assembly rotate the left cam shaft in a clockwise direction until the No. 6 exhaust valve is just closed and the inlet valve is just about to open.

3. Now mesh the cam-shaft gear in such a manner that the teeth and the flange bolt holes will line up perfectly.

The cam-shaft gear has 48 teeth and is bolted to the flange by means of seven bolts. This will permit an adjustment of one-seventh of one tooth space or $2\frac{1}{7}^\circ$ of crank-shaft travel.

4. Tighten up two of the cam-shaft gear bolts and check the tapped clearance on all left cylinders.

TO ADVANCE OR RETARD VALVE SETTING

5. Now check the opening and closing of the exhaust and inlet valves as shown on the timing diagram. If it is found that the valves are late in opening and closing, the number of degrees should be noted and the cam-shaft gear moved one or more holes in the direction of rotation without moving the cam-shaft drive shaft or the cam shaft. Remember that for each hole moved forward the cam shaft is advanced $2\frac{1}{7}^\circ$ of crank-shaft rotation. If the valves are found to open early, set the cam-shaft gear backward one or more holes.

Always check valve timing by turning engine in forward direction of rotation so as to take up all backlash in gears and lost motion in couplings.

After the gear has been properly located, set the left distributor driving flange over the bolts in such a position that the marked notch is in line with the marked tooth on the drive pinion.

Now tighten up and cotter pin the bolts and mark the gear in line with marked tooth on the drive pinion.

To set the right cam shaft, turn the engine over in the direction of rotation through 45° or until the No. 1 crank is 10° past the right dead center. With the crank shaft in this position turn the cam shaft over in a clockwise direction until the No. 1 exhaust valve is just closed and the inlet valve is just about to open. Locate the gear in the same manner as in setting the left cam shaft.

Before mounting the right distributor driving flange, turn the crank shaft back through 45° or to its original position and set the distributor drive flange so that the marked notch comes in line with the marked tooth on the drive pinion, or, in other words, in line with the center line of the right cylinders.

Now tighten up the cam-shaft gear bolts and cotter pin as before.

TIMING IGNITION

Set the two distributor assemblies in place, being careful to get them on the proper housings right and left.

These distributors are marked R and L on the outside surface of the spark-control arms. They should be fastened temporarily by means of two bolts No. 116 each in such a position that the notch on the distributor base flange coincides with the notch on the cam-shaft housing flange.

If it has been found necessary to replace either the cam-shaft housing or the distributor head, and the new parts do not carry these identifying notches, the distributor should be so set that with the spark retarded the center line of the cylinders will be midway between 1 L and 6 R terminals.

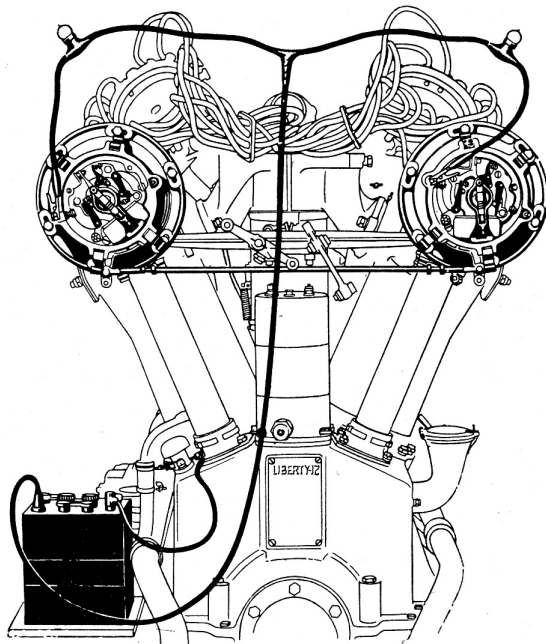
1. Set the engine on the firing point, spark retarded, No. 1 L cylinder; in other words, the neutral point of No. 6 L. Reference to the timing diagram shows that this is 10° past the top dead center.

SYNCHRONIZING BREAKERS

It is assumed that the contact separation or gap was properly adjusted (0.010 to 0.013 inch) at the time the distributor heads were overhauled. The synchronization of the two main breakers on each distributor may be checked by inserting a card between the contact points of one breaker and noting the timing of the other. Then repeating the procedure on the other breaker. If the two main breakers on a distributor do not open within 1½° of each other, they may be synchronized in the following manner: Remove the cotter pin and loosen nut on the bracket, which will allow the complete breaker-arm bracket assembly to be shifted around the stud as a center. This will vary the setting of the contact separation or gap, which should be brought back to the proper limits and the breaker arms tried again for time of opening. By shifting the breaker-arm bracket assembly and each time bringing the opening of the contact points back to the proper limits, a point can be reached where the two parallel breaker arms will open at exactly the same time, which is the proper setting for these arms. The auxiliary breaker arm must open before the parallel breaker arms when the cam is being rotated in a clockwise direction, as previously explained. If this breaker arm is not adjusted so that it will open in the above manner, it will not

prevent the back firing of the engine when being rocked on compression preparatory to starting. After the distributor breaker-arm bracket assembly has been shifted to such a position so as to give the proper opening of the parallel breaker arms, nut Q should be screwed down tight and locked in place by cotter pin P. The contact screws should be adjusted to line up and meet squarely with the contact points in the breaker arms when the contact points are closed. This can be accomplished by slightly bending and twisting the brackets which hold the contact screws.

Plate No. 66



TEST LAMPS AND CONNECTIONS

2. Swing the timing lever on the distributor to the full retarded position or as far in a clockwise direction as is possible.
3. Loosen the bolts No. 116 sufficiently that the distributor base flange can be rotated on the slotted holes.
4. Connect battery and electric light across the distributor terminals as shown on Plate 66, and rotate the distributor base flange in a counterclockwise direction until the light just goes out.

Tighten the bolts with the distributor in this position and complete the installation of the bolts No. 116.

5. Without changing the position of the crank shaft install and set the right-hand distributor in a similar manner.

6. The accuracy of the timing should now be checked up by rotating the crank shaft backward 15° or 20° ; then forward very slowly, meanwhile watching the electric lights. They should both go out at the same time within a limit of $1\frac{1}{2}^{\circ}$ on the timing disk. If the pocket flashlight is used instead of the two electric lights and battery, each distributor head will have to be checked separately and the time of the break noted according to the timing disk.

7. Install the cross reach No. 8333 and adjust it so that both distributor heads will be fully retarded. Check the synchronization of the two distributor heads with spark lever in advanced position also.

8. Install the high-tension cable tube and cable assembly, fastening it by means of the screws No. 201 to the intake headers.

9. Wire the heads of all these screws so that they will not loosen up.

Caution.—Care should be exercised in placing the distributor head assembly on the distributor to keep from breaking the rotor brush. It can best be done by putting the distributor-head assembly over the two studs upon which it is mounted and slightly rocking it back and forth with the rotor in the right-angle position to the center line of the two studs. This will gradually work the brush back into the rotor and allow the distributor head to slip down into place.

Remove wooden plugs from the spark-plug holes and install spark plugs. Connect the high-tension wires to them and to the distributors, being careful that each plug is connected to the proper terminal. Remember that plugs on the side of the cylinders toward the propeller are connected to the left-hand distributor and those on the side of the cylinders toward the gear end of the engine are connected to the right-hand distributor.

Install the water-pump assembly, being careful to see that the gasket No. 8345 is in place. Try the mesh of the water-pump gear with the water-pump bevel drive gear. This should be 0.005 inch minimum and 0.010 inch maximum. If these gears mesh too tightly, shims No. 8510 may be inserted between the bearing retainer No. 8345 and the gasket in sufficient number to produce the proper back lash.

WATER-INLET MANIFOLDS

Install the water-inlet manifolds No. 12123, replacing the hose connections and clamps wherever the original ones were found to be defective.

Put on the inlet manifold extensions No. 12096 and No. 12097 with their hose connections.

In tightening up hose clamps be careful that the clamp is bent so as to fit snugly around the hose and bear against it at all points. Do not draw up the clamps so tightly that the clamp cuts into the rubber.

It is advisable before installing the engine in the tank to tape these hose connections with a good rubber friction tape and shellac them, putting clamps on over the shellacked tape. The shellacked tape will protect the hose from oil and gasoline.

OIL-PUMP ASSEMBLY

Install the oil-pump assembly. See that the gasket No. 8348 is in good condition and is so placed that the gasket does not overlap the oil passages either in the oil pump housing or the crank case.

Install the nuts No. 101 and tighten them up evenly and cotter-pin them.

CRANK-CASE BREATHERS

Install the crank-case breather. See that the gasket No. 8172 is in place and fasten with nuts No. 101.

Install the crank-case oil fillers No. 8156 with gaskets No. 8158 and fasten in place with nuts No. 101.

Install crank-case sump cover No. 8129. Be sure that the lock No. 8130 is in its proper place.

The whole engine should now be given a careful final inspection. Make sure that all bolts have been properly drawn up and locked either by means of cotter pins or lock wires.

GOVERNOR

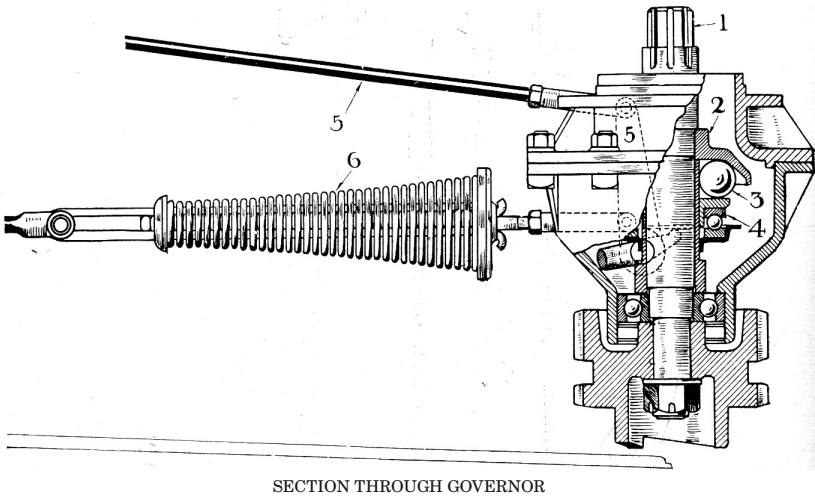
The governor is mounted on the front end of the engine, and operates on the carburetor throttle. It is designed to hold the speed of the engine below 1,400 r. p. m. A part sectional view of the governor is shown on page 111. Its method of operation is as follows:

The shaft (No. 1) rotates with the crank shaft of the engine, carrying with it the sleeve (No. 2) and the balls (No. 3). As the speed of the shaft increases the balls (No. 3) tend to fly from the center, due to the centrifugal force, moving the collar (No. 4) along the shaft and imparting a motion to the linkage (No. 5) which is connected to the carburetor throttle. The spring (No. 6) is so designed as to provide a preliminary resistance against the action of the governor, permitting the engine to reach its designed speed before the governor comes into action.

TO REMOVE ENGINE

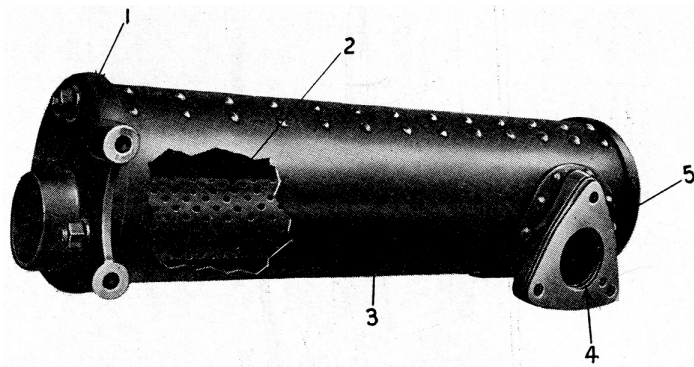
To remove the engine all the electric wires should be disconnected as well as the feed pipes and manifold connections. Disconnect the fan drive and generator drive and the connections with the cardan shaft. The plates holding the engine to its foundation should then be removed, and the engine can be lifted through the top of the hull by taking off the two bolts directly over the engine.

Plate No. 68



SECTION THROUGH GOVERNOR

Plate No. 69



MUFFLER

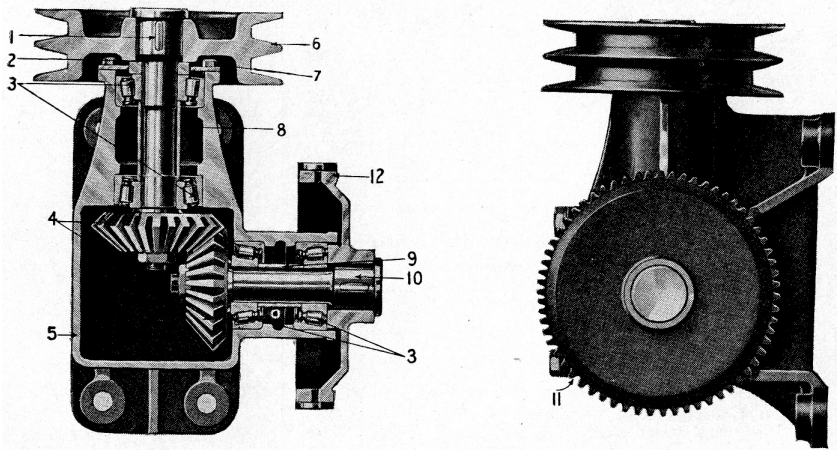
Ref. No.	Part No.	Name
1	M-39	Silencer end.
2	M-43	Silencer inlet pipe.
3	M-38	Silencer body.

Ref. No.	Part No.	Name.
4	M-48	Washer for silencer outlet flange.
5	M-41	Silencer end.

Summary of clearances

	Mini- mum	Maxi- mum	Desired
Crankshaft:			
Diametrical clearance	0.0025	0.00325	
End play	.0575	.0775	
Connecting rods:			
Forked end—			
Diametrical clearance	.003	.004	
End play	.008	.020	
Plain end—			
Diametrical clearance	.005	.0065	
End play	.004	.008	
Piston pin:			
Fit in rod	.00025	.00125	Select for 0.001 clear- ance.
Fit in piston	.00025 (loose.)	.00075 (tight.)	Select for light drive fit.
Piston rings:			
Fit in grooves	.00125	.003	Top, 0.003; middle and bottom, 0.002.
Gap	.021	.041	0.030.
Piston:			
Fit in cylinder	.018	.022	Select for 0.020 clear- ance.
Cam shaft:			
Diametrical clearance	.001	.003	
End play	.000	.004	Min., 0.002.
Cam shaft, upper drive shaft:			
Diametrical clearance—			
Large bushing	.0005	.0025	Min., 0.0015.
Small bushing	.0005	.0025	Min., 0.0015.
End play	.002	.008	Min., 0.004.
Rocker levers:			
Diametrical clearance	.00025	.00175	Min., 0.001.
End play	.005	.010	0.0075.
Valves:			
Fit of stems in guides; diametrical clearance—			
Exhaust valve	.004	.0065	0.005.
Inlet valve	.002	.0045	0.003.
Water-pump shaft:			
Diametrical clearance	.0015	.0035	Min., 0.0025.
End play	.006	.010	0.010.
Water-pump bevel driver:			
Diametrical clearance	.001	.0025	
End play	.005	.008	
Oil pump:			
Fit of gears in housing			
Diametrical clearance	.001	.005	Select for 0.004 clear- ance.
End play	.002	.007	Select for 0.003 clear- ance.
Tappet gap:			
Exhaust valve	.019	.021	
Inlet valve	.013	.016	
Breaker gap	.010	.013	
Spark-plug gap	.015	.017	0.015.
Regulator:			
Contact gap	.005	.007	
Height of pin	.043	.045	

Plate No. 70



FAN BEVEL BOX

Ref. No.	Part No.	Name
1	M-1132	Pulley spindle.
2	M-1129	Pulley distance piece.
3	M-1133	Timken roller bearing 316-312.
4	M-1125	Bevel wheel.
5	M-1123	Bevel box.
6	B-113	Driving pulley.
	21153	

Ref. No.	Part No.	Name
7	M-1126	Bearing cover plate.
8	M-1128	Long-distance tube.
9	M-1127	Short-distance tube.
10	M-1131	Chain-wheel spindle.
11	M-1124	Cover for bevel box.
12	SH193C	Chain wheel for bevel box.

Chapter IV

COMPOUND CLUTCH

Outline specifications

Type of clutch	Compound.
Slipping engagement	Asbestos cone.
Positive engagement	Driving teeth.
Cone diameter—large	19.186 inches.
Cone diameter—small	17.887 inches.
Face width of cone	2.5 inches.
Length of facing expanded	58.5 inches.
Material used in facing	Asbestos.
Material of cone	Carbon steel.
Largest diameter of flywheel	19.811 inches.
Length of clutch unit, complete	19.875 inches.
Length of cardan shaft	11.25 inches.
Diameter of cardan shaft	2 inches.
Clutch stop drum diameter	9.25 inches.

COMPOUND CLUTCH

GENERAL DESCRIPTION

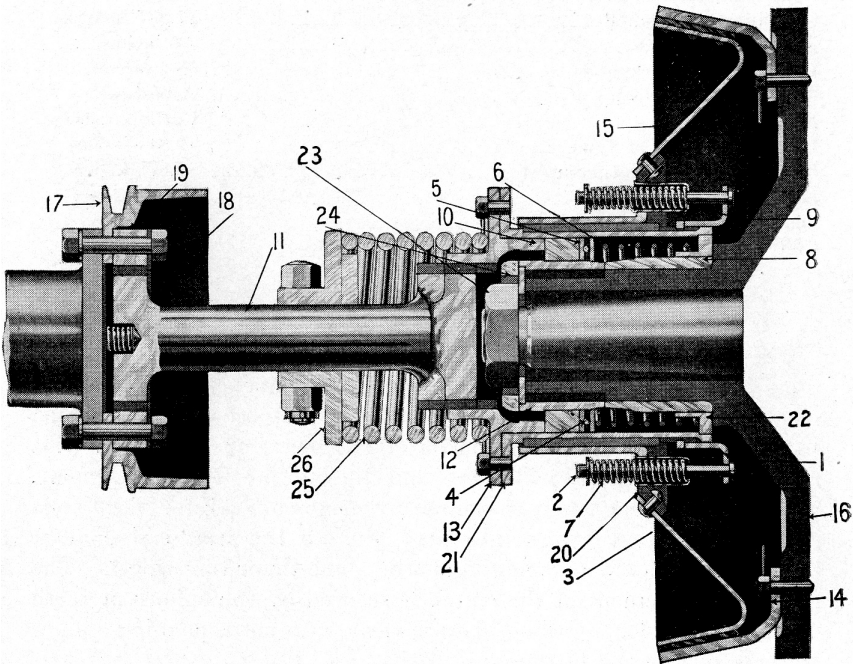
The drive is communicated from the engine to the transmission units by means of a compound clutch. Inasmuch as the complete machine weighs approximately 40 tons, the clutch has been designed to gradually overcome the inertia of the heavy mass, and for this reason provides first a slipping and then a positive engagement. The slipping engagement is secured by means of asbestos-faced cone clutch which first comes into play, due to the frictional contact between the leather facing and the cone-shaped flywheel. The positive engagement of the clutch is secured by the splines or teeth on the clutch sleeve, which in turn mesh with corresponding splines or teeth on the end of the crank shaft.

The function of the cone clutch part of the compound clutch is to simply pick up and start rotating the transmission drums. After the friction part of the clutch has accomplished this the positive drive part then is engaged. The point should be made that the epicyclic bands should and *must be* free to operate the clutch properly.

The driving-clutch unit is bolted to the flywheel member of the engine, and is a cone frustrum of carbon steel. The driving cone carrying the asbestos facing is forced into frictional contact with the driving member by means of coil springs $\frac{5}{8}$ -inch in diameter, 5.85 inches long and having 32 coils. These springs are mounted on plungers of carbon steel, $4\frac{7}{8}$ inches long, and $\frac{3}{8}$ -inch in diameter. Engagement of the clutch is secured by playing back on the clutch lever, which forces the clutch collar to slide along the clutch shaft, first engaging the cone and then engaging the splines on the collar with the splines on the crank shaft.

Additional pressure is placed upon the cone by the movement of the clutch collar by means of the large clutch spring, which is concentric upon the clutch shaft, and is composed of the coil spring, having a 5 $\frac{3}{8}$ -inch diameter of spiral and a 4 $\frac{1}{4}$ -inch length of coil. The spring is composed of five free coils, and two seating coils. Engagement between the clutch sliding collar or sleeve and the

Plate No. 71



COMPOUND CLUTCH

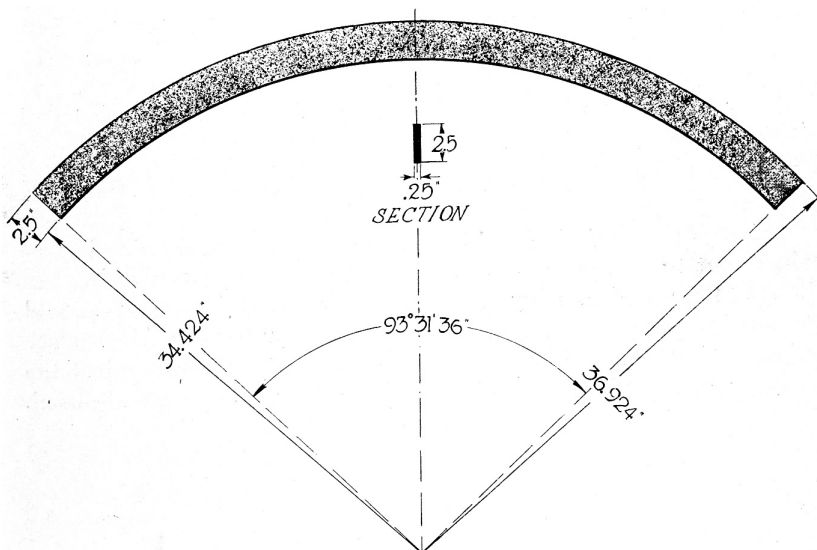
Ref. No.	Part No.	Name
1	SH861E	Clutch-spring stop ring.
2	SH861A	Spring plunger.
3	SH861C	Spring retainer.
4	SH865A	Bearing race.
5	SH865B	Ball retainer.
6	SH865C	Spring, large.
7	SH865D	Spring, small.
8	SH863A	Clutch front bearing.
9	SH863C	Clutch snap ring.
10	SH863B	Clutch rear bearing.
11	SH864A	Cardan shaft.
12	SH864C	Clutch sleeve.
13	SH864B	Clutch coupling.
14	SH866A	Outer clutch drum.

Ref. No.	Part No.	Name
15	SH867A	Inner clutch drum.
16	SH868A	Flywheel.
17	M-858	Clutch-stop drum.
18	M-856	Coupling cover.
19	M-855	Coupling box.
20	SH869A	Cone clutch-drum supporting collar.
21	SH862A	Clutch sliding collar.
22	SH862B	Clutch sliding-collar cap.
23	SH136A	Crank-shaft nut.
24	SH863D	Gear-lock washer.
25	SH849B	Clutch spring.
26	SH849A	Spring flange.

cardan shaft is secured by means of a coupling of nickel steel. The coupling is a 9-inch flange, in diameter, and a diameter of 5 inches at the point of engagement with the splined end of the cardan shaft. The cardan shaft is nickel steel, 11 $\frac{1}{4}$ inches long, with 10 splined teeth, and a diameter of 2 inches. The diameter of the splined section is enlarged to 4.359 inches.

The drive is thus imparted through the cone and through the splines on the crank shaft to the clutch collar. From the clutch collar it passes to the clutch coupling and thence by means of the splines on the coupling to the cardan shaft. The preliminary drive passes from the cone, through the carbon steel key, $4\frac{1}{4}$ inches long and $\frac{3}{4}$ inches thick and $\frac{3}{8}$ inches wide to the clutch coupling, and thence to the cardan shaft. On the rear end of the cardan shaft is located the clutch pull drum, this being bolted to the flange on the cardan shaft. A clutch brake or cardan-shaft brake operates on this drum, stopping the spinning of the cardan shaft in shifting gears.

Plate No. 72



PATTERN OF CLUTCH

It is because the cardan shaft is connected directly with the shaft of the epicyclic gear that it is very necessary that the cardan-shaft brake be properly adjusted before the machine is driven. It is quite evident that if the attempt is made to shift into reverse with this shaft spinning ahead, the driving teeth are apt to be stripped. Directions of how to adjust the cardan-shaft brake are found on page 150. The clutch-stop drum is of malleable iron.

When the clutch is thrown out of engagement the crank shaft rotates freely on the phosphor bronze clutch front bearing. This has an outside diameter of 4.684 inches, and an inside diameter of 4.002 inches. The clutch sleeve through which the main drive is transmitted is nickel steel, having a length of 2.687 inches, and inside diameter of 3.531 inches and 24 splines.

TO REMOVE CLUTCH

When removing clutch, first wire the clutch spring (exterior) to prevent it from expanding when the pressure is released. Remove the spring flange bolts.

Remove the air-pump belt from the clutch-stop drum, also the clutch-stop band pin, and the bolts connecting the band to the clutch-shaft bracket.

Remove the eight bolts holding the coupling box to the bevel-pinion shaft and the six bolts holding the clutch coupling to the clutch sliding collar.

Pull out the pin for fork ends, holding the fork end of the clutch auxiliary rod to the clutch-operating lever, enabling the clutch shaft to rotate. The cardan shaft can then be removed by dropping the rear end and allowing the ball bearings to roll clear of the shaft.

Then remove the crank-shaft nut and gear-lock washer, and pull the clutch off the crank shaft.

TO RELINE CLUTCH

Remove the old lining and chip off the rivets. New lining is provided, cut to proper size. If this is not available, a new lining can be cut to proper size, using page 117 as a pattern. New rivets should be put in in such a way that they are countersunk below the surface of the lining so that the metal of the rivet will not bear against the metal clutch-driving face, or split rivets driven from inside are used.

Chapter V

EPICYCLIC TRANSMISSION

General specifications

Type of reduction gear	Epicyclic.
Number of forward speeds	2.
Number of reverse speeds	2.
Double-gear pitch	3-4.
High-speed gear pitch	5-7.
Low-gear pitch	4-5.

GENERAL DESCRIPTION

The epicyclic gear set has two forward speeds and two reverse speeds. It is mounted transversely across the machine in the rear end of the engine compartment. It provides the necessary gear reduction between the engine drive and the track propulsion units, and also carries the drive outward on both sides from the center line of the machine to the exterior track mechanism.

The drive enters the epicyclic gear set by means of a bevel pinion drive over the engine and clutch, and leaves it in the form of a general drive to the track propulsion gearing. The gears are shifted in relation to one another by means of a shifter-fork mechanism which takes care of the forward and reverse movements, and, in addition to this, the high and low speed ratio are secured by means of brakes, which act on drums governing the ratio of the planetary trains contained in the epicyclic gear box.

The reductions in the box are 5 to 1 on low, and 1.285 to 1 on high. The reduction at the bevel gear is 14.46. The reduction between the chain sprocket and the roller pinion is 12 to 23, and between the roller pinion and the track driving wheel 9 to 37. This gives a total reduction between the engine and track of 32.545 to 1 on high speed, and 126.64 to 1 on low speed.

The use of the planetary or epicyclic type of gear box on a machine of this weight does away with the necessity of employing shifting gears except in the case of changing from forward to reverse. It is highly necessary, therefore, that the precautions mentioned on allowing all moving parts to come to rest before shifting from forward to reverse, be observed. The drive through the various units can be traced on page 123 and in the chapters dealing with drive units.

The drive from the engine enters the epicyclic gear set by means of a bevel pinion, and passes to either bevel wheel, depending on whether it is desired to run the machine backward or forward. Facing toward the front of the machine, the left bevel wheel provides a reverse motion, and the right bevel wheel gives ahead motion. Engagement

between the required bevel wheel and the shift is secured by means of a dog clutch, which slides on the splines of the bevel and epicyclic cross shaft. Both bevel wheels are in constant mesh with the bevel pinion, but only one is the driving engagement, due to the position of the clutch, the other is simply an idler, and has no part in the driving.

From the bevel wheel, in engagement with the dog clutch, the drive passes to the cross shaft, thence to the sun pinion, mounted at the opposite extremity of the cross shaft, from the bevel wheels. The sun pinion meshes with the large planet pinion, and this in turn meshes with the annular ring gear, secured to the gear casing. This casing carries with it the low-speed brake, the same bolt passing through the annular ring, and the two halves of the epicyclic casing.

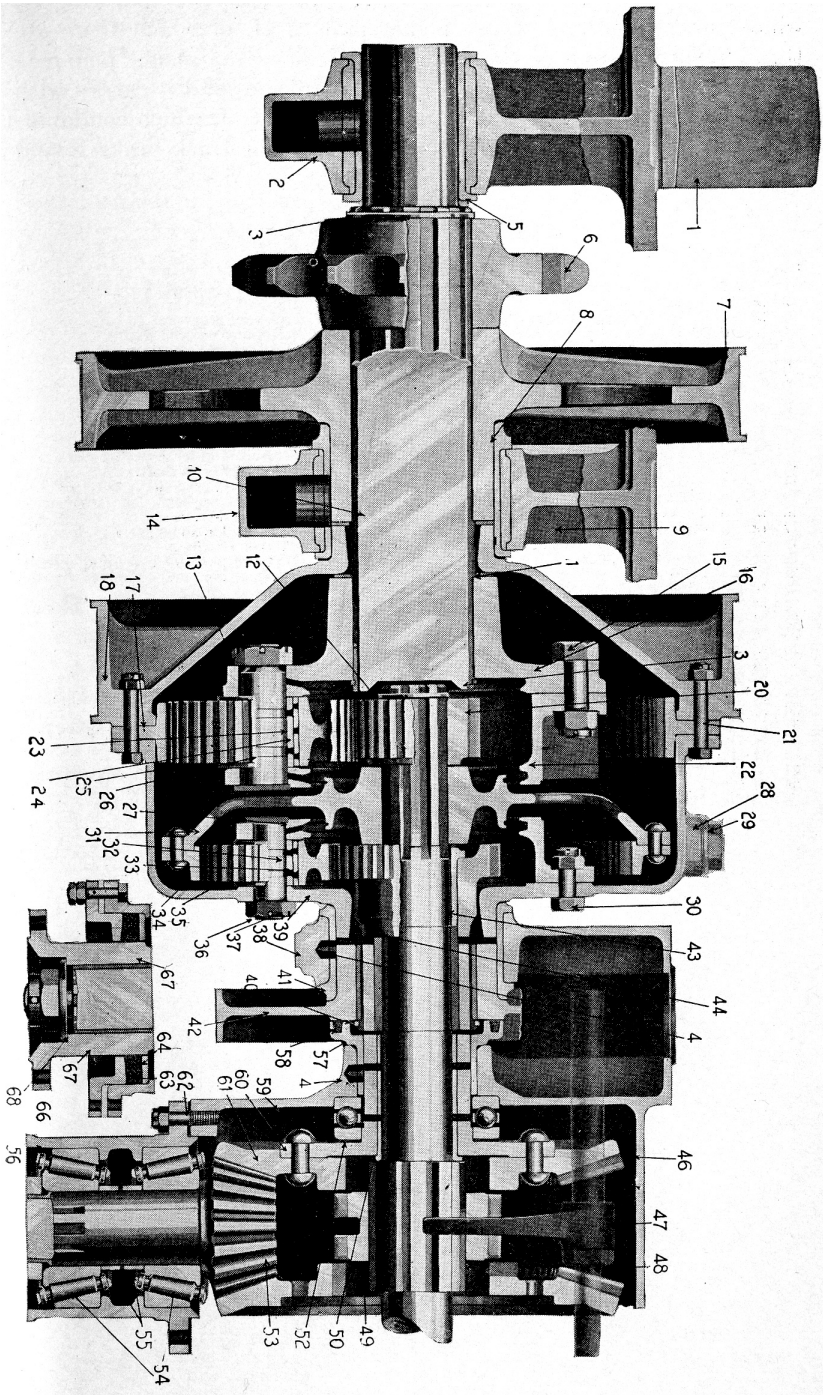
The casing also carries the drive back to the chain-drive sprocket, when it is not restrained in doing so by the application of the brake mounted upon it. The epicycle gear casing also carries a pin, upon which is mounted the small epicyclic planet pinion, which meshes with a small epicyclic spur ring, carried on an epicyclic spur disk, which in turn is splined to the epicyclic cross shaft. Revolving with the face which carries the spindle for the small epicyclic pinion is the high-speed brake, which acts as part of the gear-changing mechanism, as illustrated in plates.

PROGRESS OF EPICYCLIC DRIVE

The drive through the epicyclic gears is similar on forward and reverse speeds, except that of the bevel gears. Referring to page —, the clutch X is shifted by the fork Z so that the clutch engages with the bevel wheel B for forward speeds, and with the bevel wheel W for reverse speeds. Other than this, the progress of the drive on high and low gear is the same.

HIGH-SPEED PROGRESSION

The epicyclic transmission gears are arranged for high-speed ratio when the high-speed brake is applied on the band G, and the low-speed brake on the band N is released. Due to the position of these two brakes and the clutch X being in mesh with the bevel wheel B, the drive progresses from the bevel pinion A to the bevel wheel B, and thence through the clutch X to the cross shaft C. From the cross shaft C the drive goes to the epicyclic disk D, and thence to the spur ring V. The spur ring V meshes with the three planetary pinions E, causing them to rotate about the planet pins P, as an axis, and also to roll around the circumference of the sun pinion F. This pinion is locked in position, due to the application of the high-speed brake G. As the planet pinions E rotate about the circumference of the sun pinion F, they carry with them the pinion pins P, which in turn carry the housing Q. This housing Q is rotating about

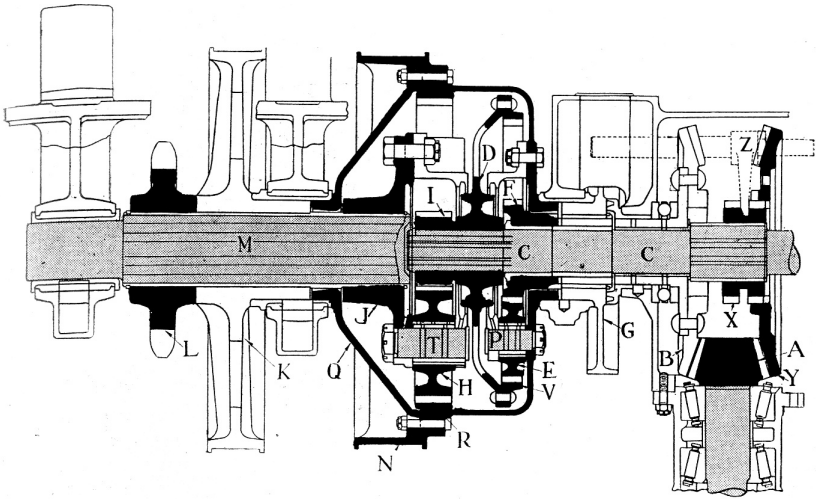


the cross shaft. The housing Q carries the epicyclic spur ring R, which meshes with the large planet pinions H, of which there are three. The rotation of the spur ring R causes the planet pinion H to roll around the sun pinion I, and as it rolls around it carries with it by means of the large planet pinion T the disk J, which communicates the drive to the shaft M and thence to the track brake K and to the chain sprocket L.

EPICYCLIC TRANSMISSION

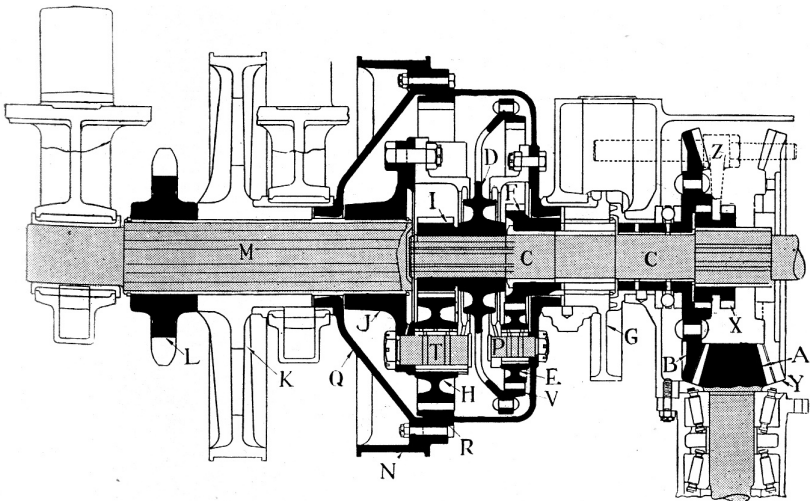
Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-297	Sprocket-bearing brackets—outside.	37	M-314	Nuts for planet pinion, small.
2	M-298	Sprocket-bearing cups.	38	M-266	Cap for brake bearing.
3	M-290	Retaining rings for shaft.	39	M-277	Epicyclic gear case.
4	M-300	Dowels for bearing bushes.	40	M-290	Retaining rings for shaft.
5	M-299	Sprocket-bearing bushes.	41	M-265	Bush for brake bearing.
6	M-291	Chain sprocket pinions.	42	M-269	Brake wheel for sun pinions.
7	M-292	Brake wheels for sprocket shaft.	43	M-268	Bush for sun pinion.
8	M-296	Sprocket-bearing bushes.	44	M-267	Epicyclic sun pinion and sleeve, small.
9	M-293	Sprocket-bearing brackets—inside.	46	M-263	Bevel-gear case.
10	M-289	Sprocket shaft.	47	M-301	Clutch fork for reversing gear.
11	M-288	Washers for planet disk.	48	M-302	Striking stud for reversing gear.
12	M-256	Retaining rings for cross shaft.	49	M-257	Clutch for cross shaft.
13	M-313	Nuts for planet pinion, large.	50	M-261	Bush for flanged sleeve, internal.
14	M-294	Sprocket-bearing caps—inside.	52	M-260	Packing washers for bevel-wheel adjustment.
15	M-318	Bolts for planet pinion ring, large.	53	M-247	Bevel pinion and shaft.
16	M-286	Disks for planet pinion, large.	54	M-281	Timken bearing.
17	M-279	Epicyclic spur rings, large.	55	M-249	Distance piece for bevel-pinion shaft.
18	M-278	Epicyclic gear case and brake.	56	M-250	Stuffing box and Timken bearing case.
20	M-287	Sun pinions, large.	57	M-262	Bush for flanged sleeve bearing, external.
21	M-315	Joint for epicyclic gear case.	58	M-310	Oil-retaining washer for bush.
22	M-285	Rings for planet pinions, large.	59	M-264	Bevel-gear case cover.
23	M-282	Bushes for planet pinions, large.	60	M-259	Flanged sleeve for bevel wheel.
24	M-283	Bushes for planet pinions, large.	61	M-258	Bevel wheel.
25	M-280	Epicyclic planet pinion, large.	62	M-254	Packing washers for bevel-pinion adjustment.
26	M-284	Pins for planet pinion, large.	63	M-253	Oil-retaining washer for stuffing box.
27	M-273	Rings for planet pinion, large.	64	M-252	Stuffing-box disk.
28	M-316	Joint ring for oil-filling cap.	66	M-251	Stuffing-box gland.
29	M-311	Oil-filling cap for gear case.	67	M-246	Flange coupling for bevel-pinion shaft.
30	M-317	Bolts for planet pinion ring, small.	68	M-248	Washer for bevel-pinion shaft.
31	M-276	Disks for epicyclic spur ring.			
32	M-272	Bushes for planet pinion, small.			
33	M-271	Bushes for planet pinion, small.			
34	M-275	Epicyclic spur rings, small.			
35	M-270	Epicyclic planet pinions, small.			
36	M-274	Pins for planet pinion, small.			

Plate No. 74



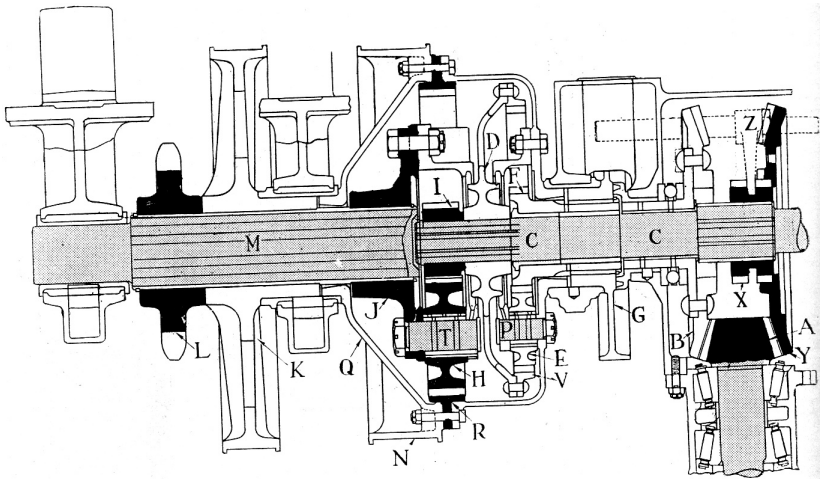
HIGH-SPEED FORWARD

Plate No. 75



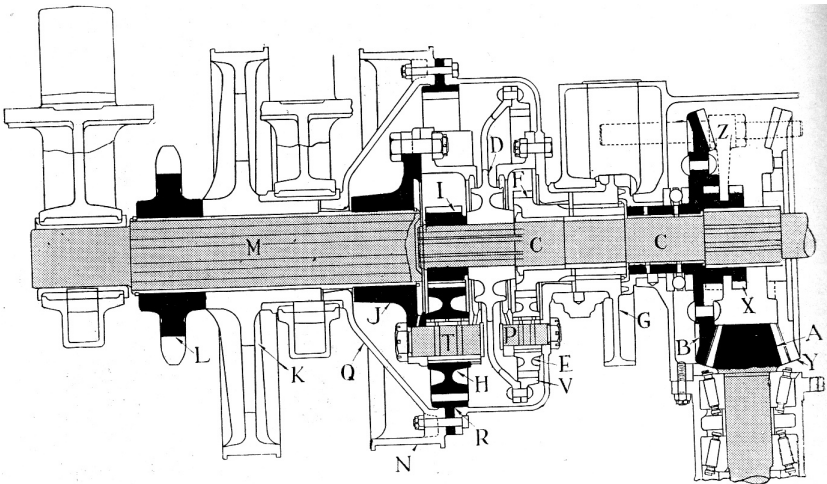
HIGH-SPEED REVERSE

Plate No. 76



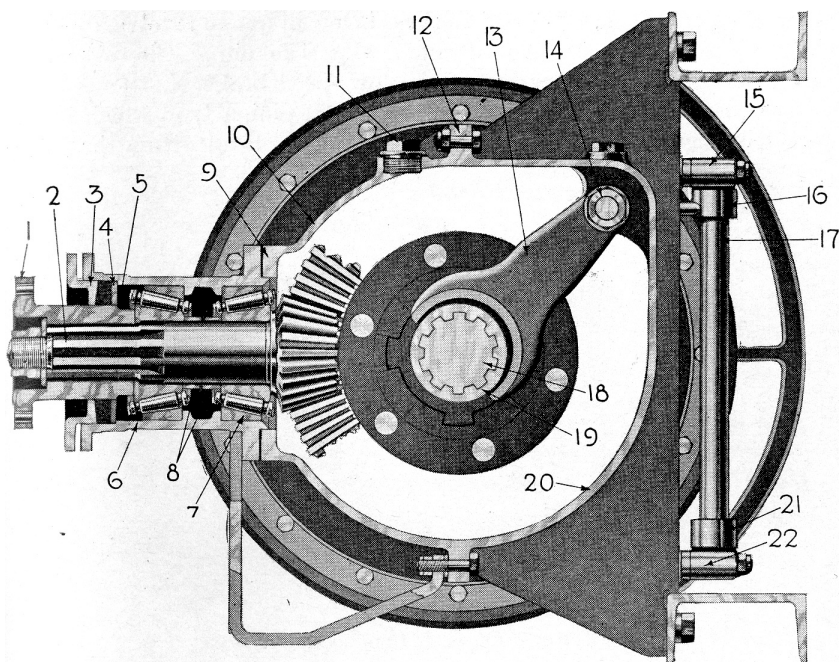
LOW-SPEED FORWARD

Plate No. 77



LOW-SPEED REVERSE

Plate No. 78



EPICYCLIC DRIVE

Ref. No.	Part No.	Name
1	M-246	Flange coupling for bevel-pinion shaft.
2	M-247	Bevel pinion and shaft.
3	M-251	Stuffing-box gland.
4	M-253	Oil-retaining washer for stuffing box.
5	M-252	Stuffing-box disk.
6	M-250	Stuffing box and Timken bearing casing.
7	M-281	Timken bearing.
8	M-249	Distance piece for bevel-pinion shaft.
9	M-254	Packing washers for bevel-pinion adjustment.
10	M-264	Bevel-gear case cover.
11	M-311	Oil-filling cap.
12	M-326	Joint for bevel-gear case.

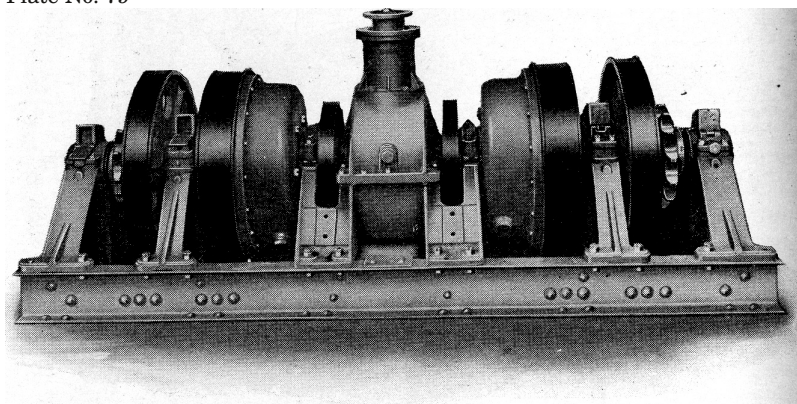
Ref. No.	Part No.	Name
13	M-301	Clutch fork for reversing gear.
14	M-302	Striking rod for reversing gear.
15	M-306	Vertical shaft bearings for reversing gear.
16	M-303	Top lever for reversing gear.
17	M-305	Vertical shaft for reversing gear.
18	M-255	Epicyclic cross shaft.
19	M-257	Clutch for cross shaft.
20	M-263	Bevel-gear case.
21	M-304	Bottom lever for reversing gear.
22	M-306	Vertical shaft bearings for reversing gear.

LOW-GEAR DRIVE

On low gear the brake is applied to the low-speed brake band, while the high-speed brake G is released. This condition of the brakes which control the speeds permits of the following progress of drive for low gear:

The bevel pinion A meshes with and drives the bevel-ring gear B, and by means of the clutch X the cross shaft C. This carries with it, idly, the parts D, V, E, F, and G, these being all free to revolve, due to the release of the high-speed brake G. The pin P, however, is fixed, due to the application of the low-speed brake N. Thus the drive goes from the cross shaft C to the sun pinion I, to which it is splined; thence to the planet pinion H, which rotates within the epi-

Plate No. 79



EXTERIOR VIEW EPICYCLIC TRANSMISSION

cyclic spur ring R, which is fixed by reason of the application of the low-speed brake. As the planet pinion H travels around within the epicyclic spur ring R it carries with the pin T and the planet disk J, by means of which the drive is communicated to the shaft M.

REVERSE SPEEDS

As stated, the progress of the drive on high reverse and low reverse is exactly the same as for high forward and low forward, except for the meshing of the clutch X with the bevel wheel Y instead of the bevel wheel B.

DRIVING-MEMBER SPECIFICATIONS

Ref. letter	Part No.	Specifications
A	M-247	14 stub teeth, 3-4 D. P.
B	M-258	46 stub teeth, 3-4, D. P.
X	M-257	4 teeth, 10 splines.
C	M-255	Epicyclic cross shaft, 10 splines.
D	M-276	30 stub teeth, 5-7 D. P.
R	M-275	78 stub teeth, 5-7 D. P.

Ref. letter	Part No.	Specifications
F	M-267	30 stub teeth, 5-7 D. P.
G	M-269	10 splines, 15 inches in diameter.
R	M-279	72 stub teeth, 4-5 D. P.
H	M-280	27 stub teeth, 4-5 D. P.
I	M-287	18 stub teeth, 4-5, D. P.

TO REMOVE THE EPICYCLIC GEAR

To remove the epicyclic gear from the hull the following operations should take place:

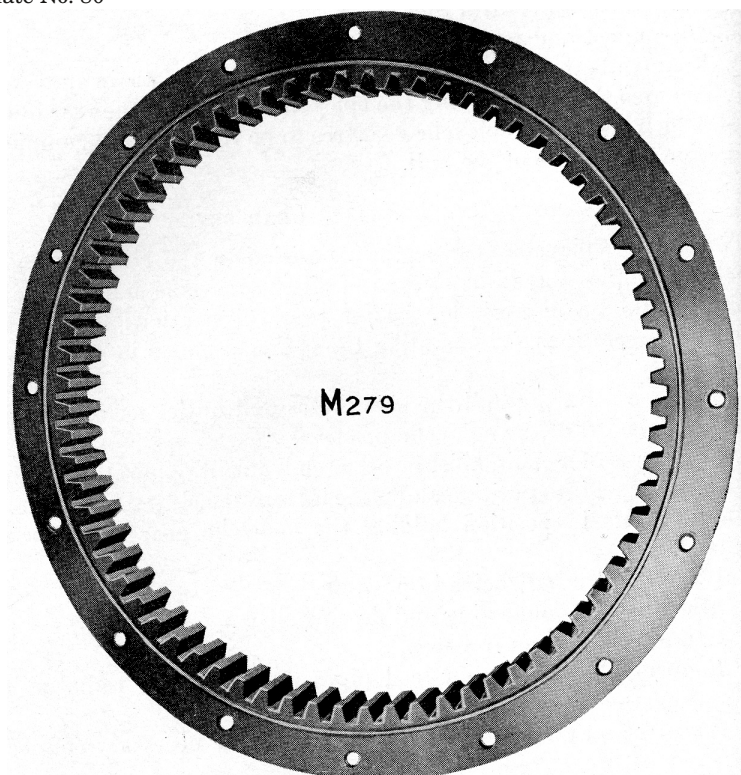
1. Remove from the hull the side plates on the right and left end of the epicyclic gear set and also the outlet louver.
2. Disconnect and remove the oil pump.
3. Disconnect and remove the gasoline-feed pipes running above the epicyclic gear set, and also remove the radiator and fan, together with the air duct.
4. Disconnect all control rods.
5. Disconnect and remove the driving shaft.
6. Remove both driving chains so as to release the chain sprocket.
7. Remove the bolts holding the epicyclic gear set to the rear bulkhead, which leaves the epicyclic gear free to be removed as a complete unit from either side of the hull.

TO MOUNT EPICYCLIC GEAR SET

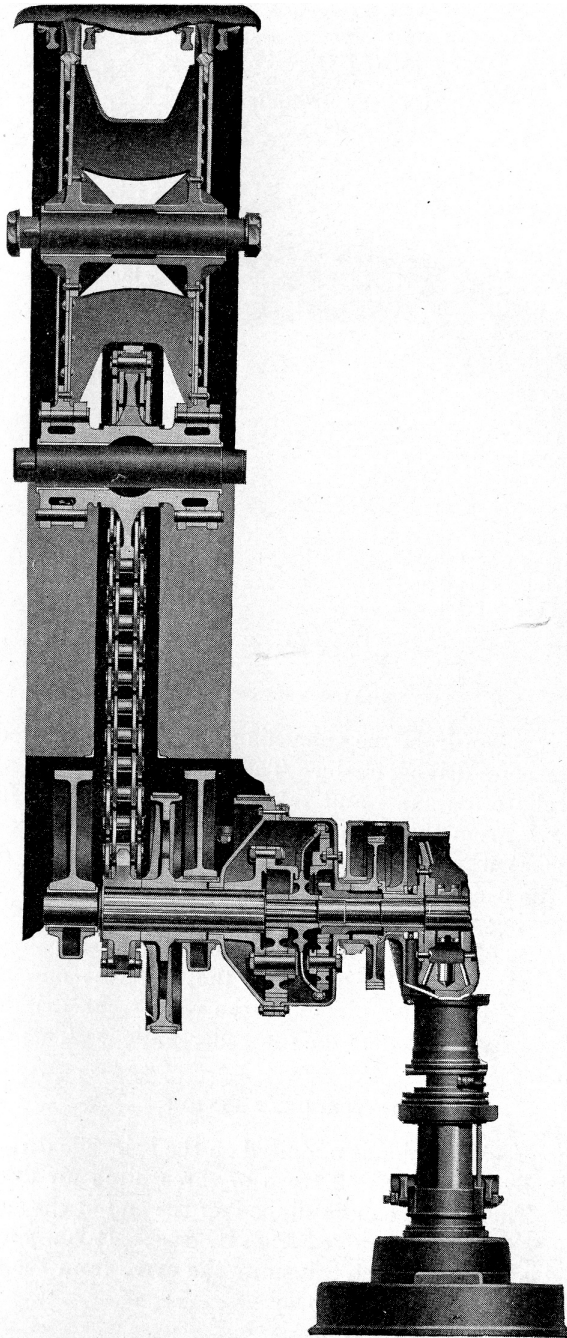
To put the epicyclic gear set into position in the hull the space which it occupies should be cleared of all obstructions and the chain casing should be in position. After these precautions have been taken the operations for mounting the epicyclic gears in position is as follows:

1. Remove from the hull the side plates and outlet louver on the right and left end of the epicyclic position.
2. By means of a chain hoist or other convenient apparatus lift the epicyclic gear set into position and place it transversely across the hull. Place the bolts in position holding the epicyclic gear to the rear bulkhead.
3. Place the driving chains in position.
4. Assemble the connecting and driving shaft.
5. Connect all the control rods.
6. Connect all the gasoline-feed pipes and place the radiator in position together with fan air duct.
7. Place the oil pump in position and make the necessary connections.
8. Replace the side plates on the hull and the outlet louver.

Plate No. 80



EPICYCLIC PLANETARY RING



PROGRESS OF DRIVE THROUGH THE VARIOUS PROPELLING UNITS OF THE 35-TON TANK, MARK VIII. THE DRIVE PASSES FROM THE CLUTCH TO THE BEVEL GEARS, THEN TO THE EPICYCLIC TRANSMISSION, FROM WHICH IT PASSES TO THE TRACK BY MEANS OF THE CHAIN, ROLLER SPROCKET, AND ROAD TRACK DRIVING WHEEL.

Chapter VI

CHAIN DRIVE

Outline specifications

Final drive	Driving sprocket chain.
Number of chains	2.
Breaking load of chains	150,000 pounds.
Maximum working load of chains	53,000 pounds.
Chain pitch	3.
Chain, inside width	1 9/16 inches.
Roller diameter	2 inches.
Maximum chain width	4 9/16 inches.
Chain length	50 pitches.
Chain pin material	Nickel steel.
Chain bushings	Open-hearth steel.
Teeth of driving sprocket	12.
Teeth of roller pinion	23.
Roller diameter	2 inches.
Roller length	2.37 inches.
Roller material	Nickel steel
Chain-sprocket diameter	23.031 inches.
Road track driving wheel diameter, outside	39.237 inches.
Road track driving wheel diameter, inside	32.75 inches.
Number of road track driving-wheel teeth	35.

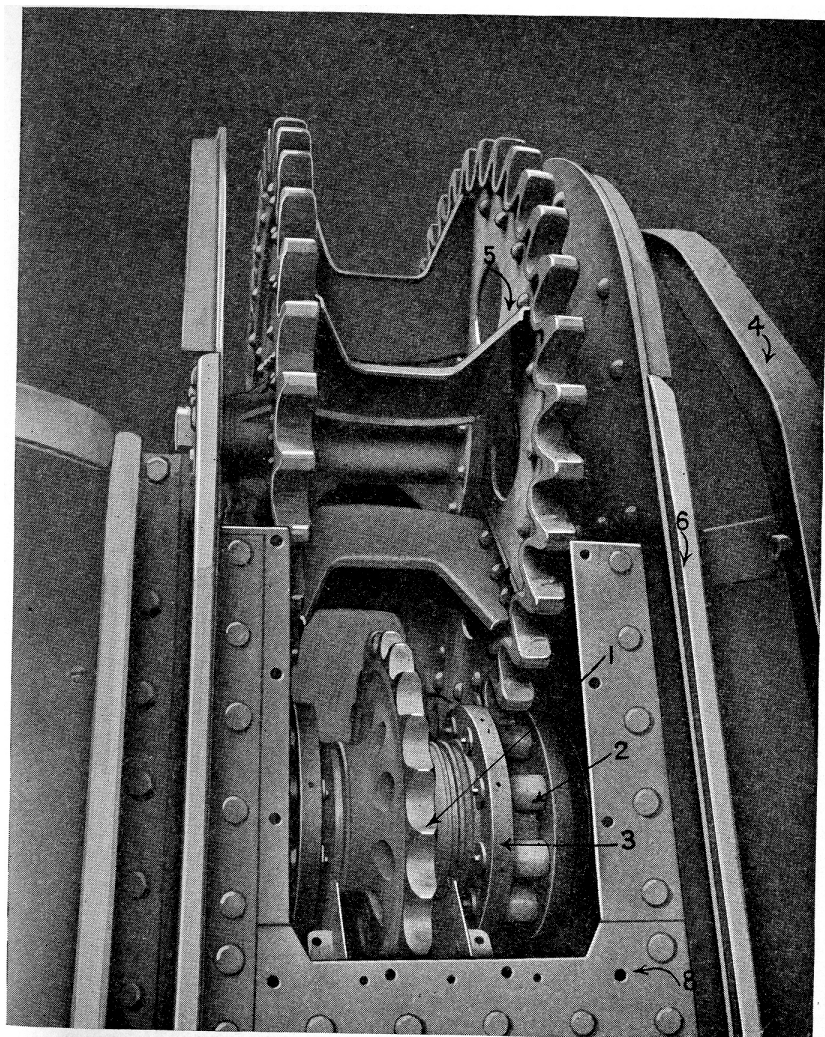
CHAIN DRIVE

GENERAL DESCRIPTION

The drive is taken from the epicyclic gear and transmitted to the track by means of a driving chain. The driving sprocket is mounted on the epicyclic cross shaft and takes the drive directly from the epicyclic gear. From this point it passes to the roller pinion which has the sprocket wheel at its center and the roller-pinion drive on each side of the center. These roller pinions mesh in turn with the road track driving wheel, which engages directly with the track; so that the progress of the drive is from the epicyclic gears through the driving sprocket; thence by chain to the sprocket on the roller pinion, and from the roller pinion to the road track driving wheel which is driven by the rollers on the roller pinion. This drives in turn the track.

CHAIN-SPROCKET PINION

The chain-sprocket pinion is mounted on the epicyclic driving shaft and is driven by it. It has 12 teeth of three pitch for driving the roller chain. This pinion can be slipped off the end of the shaft after the epicyclic assembly is removed from its case. It is a part of this assembly and is the unit which transmits the drive from the epicyclic



ROLLER PINION AND TRACK DRIVE

Ref. No.	Part No.	Name
1	M-1541B	Chain sprocket and roller pinion.
2	M-1542	Roller.
	M-1543	Pin for roller-pinion rollers.
	M-1550	1/8-inch taper plug for roller pin.
	M-1409	Driving wheel.
	M-1544	Shaft for roller pinion.
	M-1407	Outer shaft bearing.
	M-1546	Inner shaft bearing.

Ref. No.	Part No.	Name
	M-1552	Plate for shaft bearing.
	20297-D89	Wheel shaft.
	M-1549	3/4-inch taper plug in shaft.
	M-1541A	Roller-pinion sprocket.
4		Guide rail.
5		Road track driving wheel.
6		Upper track rail.

gearset to the chain. The chain sprocket pinion is a drop forging and engages with the epicyclic shaft by means of 10 splines. It has a total width on the shaft of 4 inches. Its pitch is 11.592-inch and its teeth are $1\frac{1}{2}$ inches wide.

DRIVING SPROCKET CHAIN

The driving sprocket chain is a chain which connects the chain sprocket pinion on the epicyclic transmission system with the roller pinion which in turn drives the track driving wheel. The driving sprocket chain is designed to operate under a maximum working load of 53,000 pounds and has a breaking load of 150,000 pounds. It is a 3-pitch chain, and has a length of 50 pitches. The chain is made up of outer and inner bars which connect the rollers. The outside and inside bars are of 40 carbon, hot-rolled steel. The distance between the outside bars is $2\frac{7}{8}$ inches and between the inside bars $1\frac{9}{16}$ inches. The outside bars are $\frac{1}{2}$ inch thick and the inside bars are $\frac{5}{8}$ inch thick. The pins are $1\frac{7}{32}$ inches in diameter, and are of nickel steel. The pins are held in place by a $\frac{3}{8}$ inch by $1\frac{3}{4}$ inch split pin. The holes through which the pins pass are reamed to 1.231 inches, giving a 0.005 clearance between the pin and the hole. Surrounding each pin is an open-hearth steel bushing which takes the wear.

CHAIN SPROCKET AND ROLLER PINION

The chain sprocket and roller pinion is one assembly, being cast iron and carrying the roller sprocket in the center, and the roller pinions on each side of the center. The drive is transmitted by the chain to the chain sprocket, which is an iron casting, also carrying the roller pinions shaft which is fixed and screwed to the inside and outside plating of the hull. The chain case surrounds the central portion of the casing, or that portion that is known as the chain sprocket.

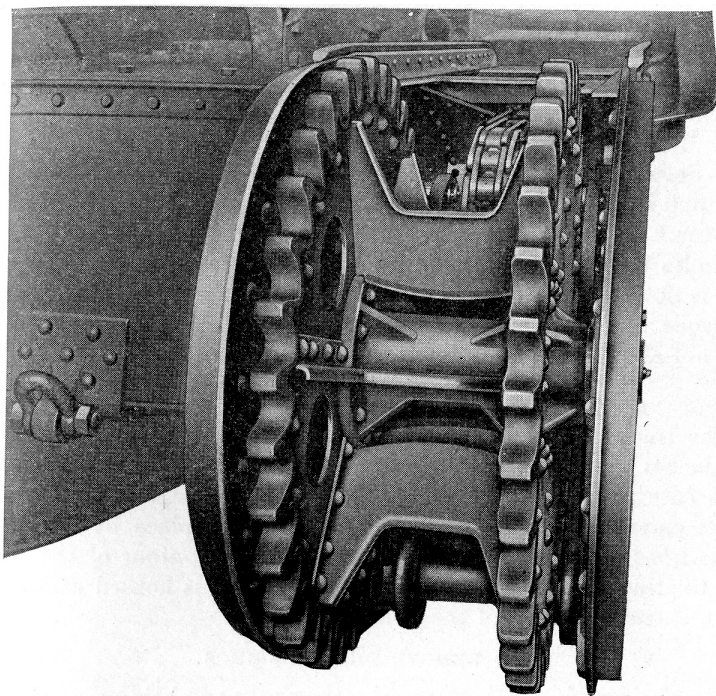
The length over all of this unit casting which acts as a chain sprocket and a carrying part for the roller pinion is $19\frac{1}{2}$ inches. The diameter at the roller-pin bosses is 13.687 inches, and the diameter of the sprocket is 23.031 inches. The sprocket has 23 teeth, 1.39 inches wide. The rollers are nickel steel, 2 inches in diameter, and 2.37 inches long. They revolve on pins, passing between the bosses, these pins being nickel steel 5.593 inches long and $1\frac{3}{4}$ inches in diameter. The shaft about which the roller pinion revolves is carbon steel $2\frac{1}{4}$ inches long and 4.434 inches in diameter.

The casting revolves on the bronze bushing, and is lubricated by the hole drilled through the center of the shaft, and having leads from the center out to the bushing. The ends of the hole through the center of the shaft are sealed by $\frac{3}{4}$ inch pipe plugs, the one on the inside being cut off flush after being screwed in, and the one on the outside being left with its head so that it can be removed for the purpose of introducing oil.

ROAD TRACK DRIVING WHEEL

The roller pinions mesh with the road track driving wheel. This is of cast manganese steel, 39.237 inches in diameter, having 35 teeth, 3.735 pitch and 2 inches in width. The tooth operation of the steel is a manganese steel casting, these two tooth rings are riveted to a series of diaphragm plates which are in turn riveted to the road track driving wheel disks. The disks are riveted to bosses of the

Plate No. 83



ROAD TRACK DRIVING WHEEL

Part No.	Name	Part No.	Name
M-1401	Teeth for road track wheel.	M-1409	Bushing for road track wheel.
M-1402	Shaft for road track wheel.	M-1410	Plug in end of shaft.
M-1403	Boss for road track wheel.	M-1411	Locking plate for shaft nut.
M-1404	Disk for road track wheel.	M-1552	Stud plate for shaft bearing.
M-1405	Diaphragm for road track wheel.	20297-D89	Key for road track driving wheel shaft.
M-1406	Shaft bearing inner end.	M-1477	Nut for shaft.
M-1407	Shaft bearing outer end.		

road track driving wheel, and this entire assembly revolves on the road track driving-wheel shaft bronze bushing. This bushing is lubricated similar to the bushing of the roller pinion and chain sprocket by means of a lead drilled through the axis of the driving-wheel shaft to about the center of its length. From this point there is a lead to the bushing. The opening to the oil lead is sealed by a pipe plug which is removed when oil is introduced.

The road track driving-wheel shaft is carbon steel, 27.625 inches long and 4.434 inches in diameter. The bushing upon which the road track driving wheel rotates is bronze, 5.118 inches in diameter, and 8 inches long, and interchangeable with the bushing upon which the chain sprocket and roller pinion rotates. This bushing is renewable after wear by simply driving them off the shafts and replacing them.

TO REMOVE THE DRIVE CHAIN

To remove the drive chain, the following operations should be followed in the order given:

1. Break the track over the rear mud pocket by removing one of the connecting links at this point.
2. Roll drive chain around until the key link is on the top.
3. Remove the key link.
4. Secure hold on the end of the chain coming from the under side and pull it through.

NOTE.—It must be remembered in replacing the chain that the oil holes must be on top and the cotter pins on the outside. If this is not done the cotter pins will catch on the gusset plate of the epicyclic which have been cut away for the chain links, but which will not allow the cotter pins to pass by freely.

CHAIN CASING

The track driving chain is surrounded by a sheet steel casing, 6 $\frac{5}{8}$ inches wide and extending around the chain for its entire length. This case is supported from the outside and inside plating of the track extensions of the hull. It is bolted in place by means of angle irons, and is so shaped as to follow the contour of the chain and to clear the web on the track sprocket. It is housed within the track-extension plates of the hull.

TO REMOVE CHAIN CASINGS

The chain casing is in two parts, the outer end being a removable cap fitting over the sprocket of the roller pinion. In removing the chain casing the following operations should be followed:

1. Remove track over the roller pinion.
2. Remove the bolts from the angle-iron cleats on both sides of the cap.
3. Remove the cap.

In removing the main casing, it is necessary to remove the track driving wheel, roller pinion, driving chain, and epicyclic gear. After these are taken out, the nuts fastening the chain casing to the back-plates of the hull and the supporting bracket must be removed. The casing can then be taken out.

NOTE.—In reassembling, the chain casing must be fixed in position before the epicyclic gear is mounted.

TO REMOVE ROLLER PINION

To remove the roller pinion, the following operations should be followed:

1. Break the track and remove the drive wheel.
2. Unbolt the chain casing and disconnect and remove the drive chain.
3. Remove the nuts holding the shaft bearing (outer end).
4. Insert screws in the tap holes in the shaft bearing at the outer end and screw them, forcing the shaft bearing out.
5. After the shaft bearing has been removed, the shaft can be drawn out and the roller pinion removed. This also permits the floating bushing to be taken out.

Chapter VII

ROAD TRACK AND TRACK DRIVE

Outline specifications

Type of track	Continuous, linked.
Length of ground contact (normal)	9 feet 3½ inches.
Length of track expanded	72 feet 6 inches.
Number of track links	78.
Road track drive	Roller sprocket.
Track shoe material	Armor steel.
Diameter of track adjusting steel	40.187 inches.
Diameter of road track driving wheel	23.031.

GENERAL DESCRIPTION

The road track passes in the form of a continuous belt, running around the entire exterior of the hull. There are two of these tracks, one on each side of the machine. The tracks are propelled by means of a road-track driving wheel which carrying the sprocket engages with the pins on the track links. As the track is driven around the hull, it carries with it the track links which lay down before the machine a railbed, over which the machine rolls, by means of the road-track rollers.

These road-track rollers run on the rail faces of the track links, giving a continuous rail surface, regardless of road conditions. Thus the road-track driving wheel through its mesh with the road-track chain, not only propels the machine but also supplies the continuous surface necessary for the rollers.

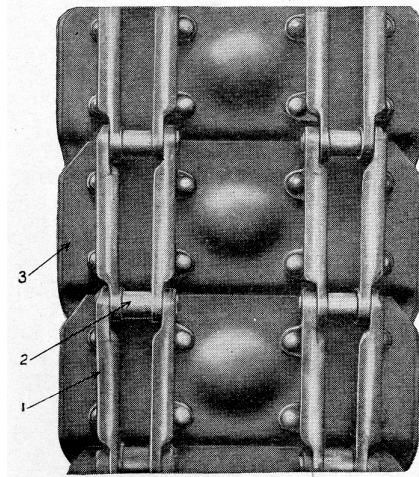
The road-track driving wheel is at the rear of the machine. At the forward end of the machine there is an idler wheel, which serves as a guide for the track, and at the same time provides a means of adjustment. The position of the idler wheel fore and aft determines the center to center distance between the idler and adjusting wheel, and the sprocket, and hence governs the tension on the track. Moving the road track adjusting wheel forward tightens the chain, and moving it toward the rear loosens the chain.

There are 58 rollers which carry the weight of the machine. These are known as lower rollers, as distinguished from the top road track rollers, of which there are two. Of the 58 rollers, 28 are fitted with spring plates which act as spacers, and 30 are plain without the spring plates. Since the road track is double, these spacers keep the rollers which roll upon the track at the proper distance apart. The two top road-track rollers are alike, one on each side at the point where the track makes its sharpest angle and where, consequently, the greatest strains may be expected. Elsewhere along the top the track slides on the top track rails.

TRACK LINKS

The track links are drop forged steel, having a length from pin center to pin center of 11.154 inches. The assembly of these links must be very accurate, and the allowable limit in assembly is 0.004 inches between pin centers; i. e., when links are assembled in pairs they must agree to within 0.004 inches between pin centers. The links perform the double duty of binding the chain together, and also of forming a rail surface upon which the machine rolls. It takes a left and right link bar to make up one link assembly. These bars are connected by the pins which are driven through. The pins are sur-

Plate No. 84



ROAD-TRACK LINK

Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1265	Road-track link.	3	1264	Road-track shoe.
2	M-1263	Bushing for road-track link.			

rounded by carbon-steel bushings, which take the wearing stresses, due to meshing with the road-track driving wheel. Riveted to the links are the track shoes, which are in contact with the ground. The track shoes are pressed from armor plate, and so shaped that they link one over the other. The shape of the track shoe, and the method, of overlapping is clearly shown above. The track shoe is 26½ inches in width. Under normal conditions there is a total area of 41.052 square feet of track surface in contact with the ground.

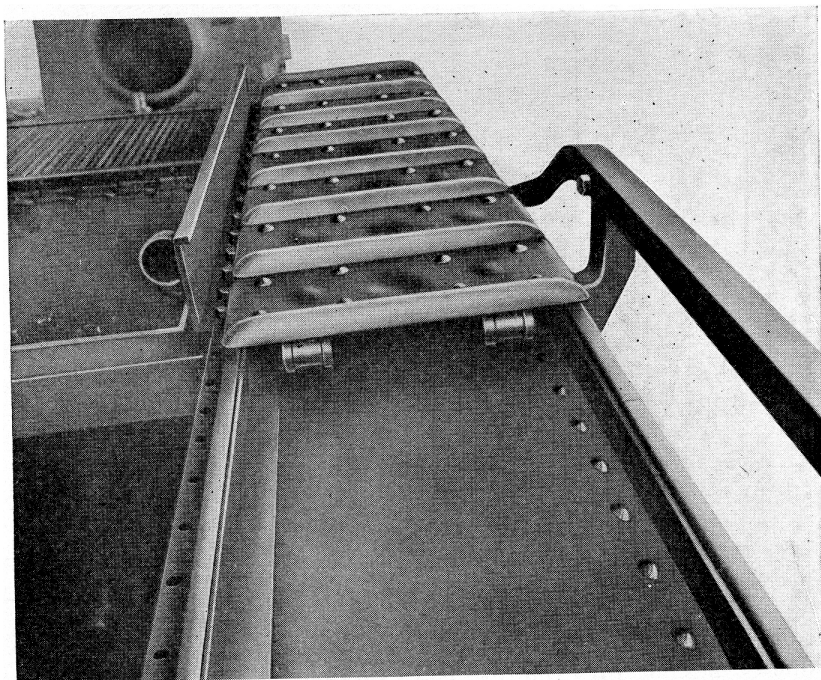
ROAD-TRACK DRIVING WHEEL

The road-track driving wheel has been described in Chapter VI, see page 133.

ROAD-TRACK ADJUSTING WHEEL

The road-track adjusting wheel is an idler wheel, having no teeth and running idly with the track. Its sole function is to act as a guide for the front end of the track, and also act as a track adjustment, by means of the belt which allows it to be moved lengthwise, thus increasing or decreasing its distance from the road-track driving wheel. The road-track adjusting wheel has a diameter of 40.187 inches. It is of diaphragm construction, the units composing it being a solid rim which runs upon the track surface of the links. The disc connects the diaphragm with the rim and with the boss which acts as a hub.

Plate No. 85



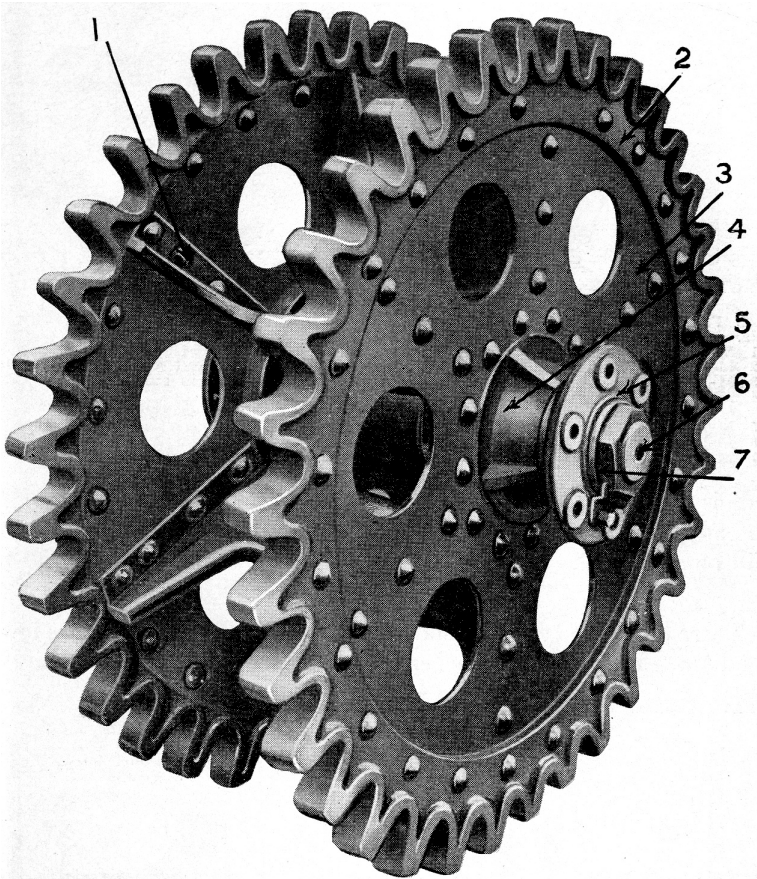
TOP TRACK RAIL UPON WHICH TRACK SLIDES. TRACK SHOWN BROKEN

These combined units make up the road-track adjusting wheel proper and in addition there is the adjustment feature, consisting of an adjusting screw which moves the entire action of the adjusting wheel forward or backward as desired. In order to preserve the alignment of the wheel there is one of these adjusting screws on each side of each adjusting wheel. The tension adjusting screw is nickel steel, 15 $\frac{3}{8}$ inches long, 1 $\frac{1}{8}$ inches in diameter. It is located within a bracket of cast steel, and there is a guard in the hull plate for the track-adjusting screw. This guard or opening is 6 $\frac{3}{4}$ inches long and 4 $\frac{3}{4}$ inches wide.

TRACK-ADJUSTING SHAFT

The track-adjusting shaft is surrounded by a bronze bushing upon which the track-adjusting wheel rotates. This bushing is interchangeable with the bronze bushing on the track driving wheel and

Plate No. 86



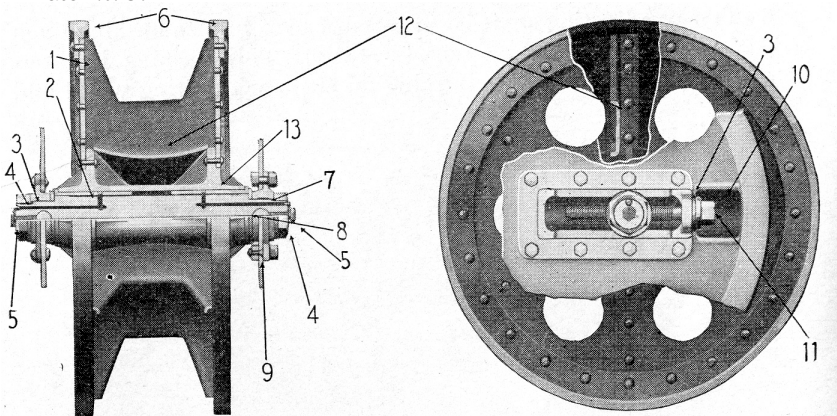
ROAD-TRACK DRIVING WHEEL

Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1411	Locking Plate for shaft nut.	8	M-1410	Plug in end of shaft.
2	M-1401	Teeth for road-track wheel.	9	2097-D89	Key for road track driving-wheel shaft.
3	M-1405 xy	Diaphragm road-track wheel.	10	M-1409	Bush for road-track driving wheel.
4	M-1404	Disc for road-track wheel.	11	M-1406	Shaft bearing, inner end.
5	M-1403	Boss plate for road-track wheel.	12	M-1402	Shaft for road track wheel.
6	M-1552	Stud plate for shaft bearing.	13	M-1477	Nut for shaft.
7	M-1407	Shaft bearing, outer end.			

on the roller sprocket. The track-adjusting shaft is drilled for oil passages, the holes being sealed by pipe plugs, which are removed for the purpose of injecting oil. These plugs are $\frac{3}{8}$ -inch pipe plugs.

The nuts of the track-adjusting shafts are hexagonal, $4\frac{1}{8}$ inches in diameter and 4.18 inches across flats. Inserted in the shaft on the

Plate No. 87



ROAD-TRACK ADJUSTING WHEEL

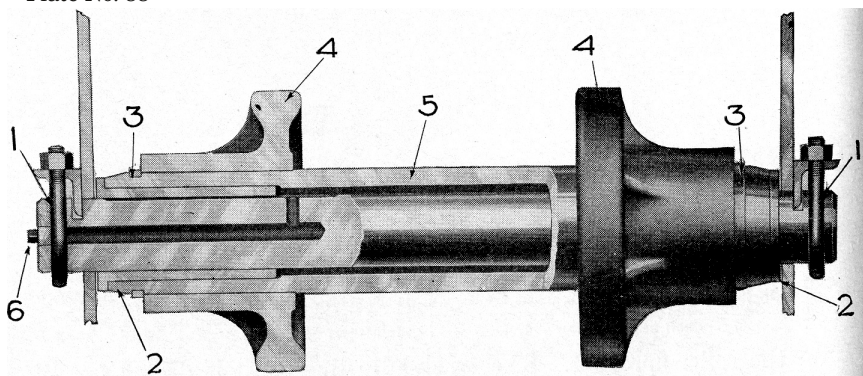
Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1404	Disc for road-track adjusting wheel.	8	M-1474	Shaft for tension-adjusting screw.
2	M-1409	Bush for road-track adjusting wheel.	9	M-1484	Plate for track-adjusting bracket.
3	M-1472	Bracket for tension-adjusting screw.	10	M-1479	Guard in hull plate for adjusting screw.
4	M-1477	Nut for shaft for track-adjusting wheel.	11	M-1473	Tension-adjusting screw.
5	M-1475	Locking screw for adjusting wheel.	12	M-1405	Diaphragm road-track adjusting wheel.
6	M-1471	Rim for road track.	13	M-1403	Boss for road-track adjusting wheel.
7	M-1472	Bracket for tension-adjusting screw.			

side is a hard copper plug for locking the track-adjusting screw. As the adjusting screw is turned it moves its entire shaft lengthwise, as will be explained.

ROAD-TRACK ROLLERS

There are three types of road-track rollers. Two of these are lower road-track rollers and the other is the top or upper rollers. The two

Plate No. 88



TOP ROAD-TRACK ROLLER

Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1338	U bolt.	4	M-1341	Top roller.
2	M-1339	Floating bushing.	5	M-1333	Tube.
3	M-1335	Spring ring.	6	M-1410	Tap plug in roller pin for oil.

types of lower road-track rollers are similar with the exception that one has spring plates and the other has not. The three types of rollers are shown on pages 140 and 143. The purpose of the lower road-track roller is to carry the weight of the machine and provide the wearing surface which bears upon the tracks, held down by the road-track links. The machine rolls on this surface upon the steel roller provided by the track linkage.

LOWER ROAD-TRACK ROLLERS WITH SPRING PLATES

The road-track rollers are mounted between the outer and inner shell plating of the hull and are carried on shafts which are secured in position by means of angle brackets and U bolts which suspend the shafts and hold them in position. The shafts are shaped to fit the toe of the angle bracket, thus locking them securely against turning, as well as providing a clearance for the tracks. (Page 143.) The distance between the inner and outer shell plating along the line of the lower road-track roller is $22\frac{1}{4}$ inches. The rollers themselves roll upon the two double tracks provided by the double set of links in the track chain.

The rollers are $8\frac{3}{4}$ inches in diameter and are $4\frac{7}{8}$ inches wide. Holding these rollers in their proper relationship on the shaft, on 28 of the lower road-track rollers there are spring plates. Between these plates are the road-track-roller springs, 1 inch in diameter, having three coils 6.652 inches long under a load of 2,500 pounds. The spring has an outside diameter of 3.937 inches.

The road-track-roller pin which acts as a shaft for the rollers extends entirely across the housing formed by the inner and outer shell plates and is $25\frac{3}{4}$ inches in length and 2.247 inches in diameter. At its inner and outer ends it carries a bronze bushing, known as the road-track-roller tube bushing, upon which the road-track-roller tube together with the two road-track rollers, rotate.

The road-track-roller pin is drilled partially at each end, providing an oil passage, and is also drilled radially to provide an oil passage to the bushing. The road-track-roller tube or sleeve is $21\frac{1}{2}$ inches long, 3.812 inches outside diameter, and 2.75 inches inside diameter. It carries the rollers, the spring plates, and the spring, all of which revolve as a unit, together with the tube. The rollers are locked to the tube by means of a spring-steel split ring of carbon steel, 0.187 inch thick, 0.375 inch wide, and 3.531 inches in diameter.

LOWER ROAD-TRACK ROLLERS WITHOUT SPRING PLATES

The lower road-track-rollers without spring plates are 30 in number and are similar to the lower road-track rollers with spring plates in all respects except that the plates and spring are omitted. These rollers are placed alternately with the spring rollers.

TOP ROAD-TRACK ROLLERS

There are two top road-track rollers, located one on each side of the machine at the top, at the point where the track has an angle, due to the contour of the machine. These road-track rollers are similar in most respects to the lower road track rollers, without spring plates. The rollers are cast iron, $8\frac{3}{4}$ inches in diameter, operating on the tube or cylinder in the same way as the lower road-track rollers, the sleeve being interchangeable. The top road-track-roller pin is also interchangeable with the lower road-track-roller pin, and the assembly is quite similar throughout, except for the shape of the roller.

TO REMOVE THE TRACK

To remove the track, slacken the locking screw and the track-adjusting screws. These are the small nuts in the center of the nut on the end of the track-adjusting wheel shaft. Also slacken the nuts for the shaft of the track-adjusting wheel on each side of the track. This allows the following operations to be carried out in removing the track:

1. Unscrew the adjusting screws on each side of the track-adjusting wheel until there is considerable slack.

2. Raise the track links over the mud chute until the road-track link pins are clear of the bulb rail upon which the track slides at the top of the machine. Pull the two links together as far as possible to reduce the pressure on the link pins.

3. Remove the road-track-link cotter pins and drive out the link pin with a drift.

4. Remove the track from the top by pinching the bars or lifting the ends by means of a crane.

The machine is then jacked up and the track removed from beneath.

TO REMOVE TRACK LINK

If it is necessary to remove a link in any part of the track, the easiest way is to break the track over the mud pocket, in order to get plenty of slack where the link is to be taken out, rather than to have the slack obtained merely by releasing the tension of the track-adjusting screw. This will greatly simplify the removal of the link.

TO ADJUST THE TRACK

The track adjustment is accomplished by moving the road-track-adjusting wheel forward to increase the tension on the track, or backward to decrease the tension. In other words, moving the adjusting wheel forward is increasing the center distance between the road-track wheels, thereby drawing the track tighter. This is accomplished by slackening the locking screw for the track-adjusting screw; that is, the small screw in the center of the nut on the end of the shaft

of track-adjusting wheel. The adjustment is then made by screwing up on the tension-adjusting screw until the desired tension is obtained.

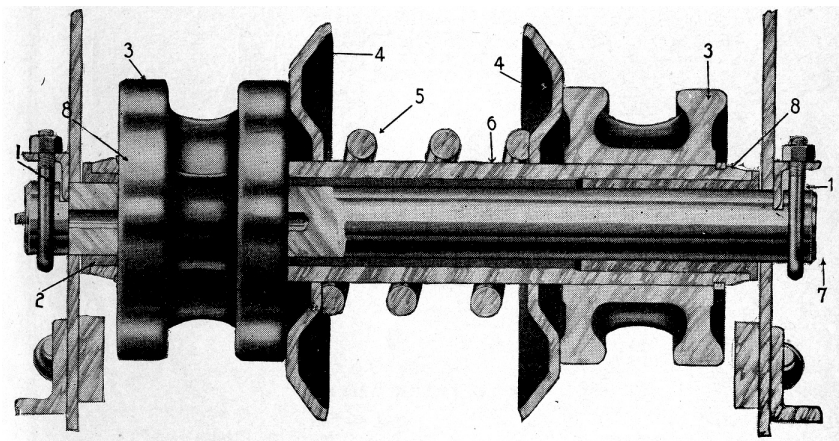
TO REMOVE THE ROAD-TRACK-ADJUSTING WHEEL

After removing the track, slacken the locking screws and remove the nuts from both ends of the shaft. Unscrew both tension-adjusting screws until clear of the shaft. Remove the outside track-adjusting bracket and drive the shaft through the opening. The adjusting wheel should be blocked up before drawing the shaft.

TO REMOVE THE ROAD TRACK DRIVING WHEEL

After removing the track, block up the driving wheel. Remove the locking plates for shaft ends on both sides of the wheel. Unscrew the

Plate No. 89



LOWER ROAD-TRACK ROLLER WITH SPRING

Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1338	U bolt.	5	M-1336	Spring.
2	M-1339	Floating bushings.	6	M-1333	Tube.
3	M-1332	Roller.	7	M-1337	Roller pin.
4	M-1334	Spring plate.	8	M-1335	Spring ring.

track driving wheel nuts. Remove the bolts holding the shaft bearing (outer end).

Insert screws in the tap holes in the shaft bearing (outer end) and screw them in, forcing the shaft bearing out. After removing the outer shaft bearing the shaft can readily be pushed out from the inner end.

The driving wheel is then removed and the floating bushing taken out.

TO REMOVE TOP ROAD TRACK ROLLER

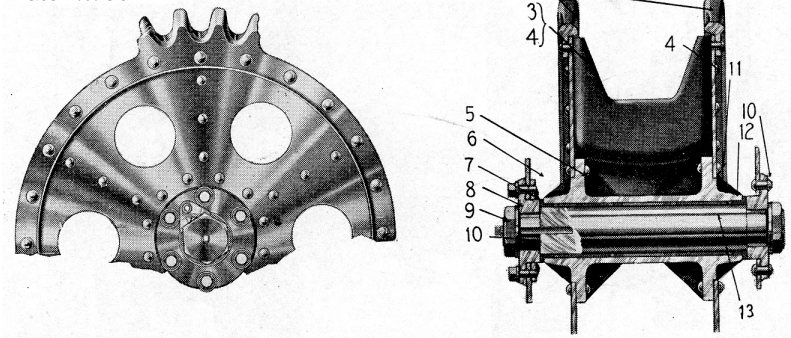
After removing the track, remove the nuts from the road track roller pin staple, or U bolt, removing the staple. Remove the bolts on the roller support angles and remove the angles themselves, after

which the road track roller pin can be forced out the outer side of the hull. The road track rollers can then be removed and the floating bushings taken out. The inner end of the rollers can be reached through the engine room.

TO REMOVE LOWER ROAD TRACK ROLLERS

Jack up the machine from the ground by placing jacks so as to put the weight of the machine on the hull plating. Then remove the track in accordance with the directions given on page 142. Remove the nuts on the road track roller pin staple and take off the staple. Unscrew the plates on the roller support angle and remove them, after which the lower road track roller pin can be driven out of the end. The road track roller can then be removed and the floating bushing taken out.

Plate No. 90



ROAD TRACK WHEEL

Ref. No.	Part No.	Name
1	M-1405xy	Diaphragm road track wheel.
2	M-1401	Teeth for road track wheel.
3	M-1404	Disk for road track wheel.
4	M-1403	Boss for road track wheel.

Ref. No.	Part No.	Name
5	M-1407	Shaft bearing, outer end.
6	M-1402	Shaft for road track wheel.
7	M-1477	Nut for shaft.
8	M-1411	Locking plate for shaft nut.

TO REMOVE ROLLER SPRING

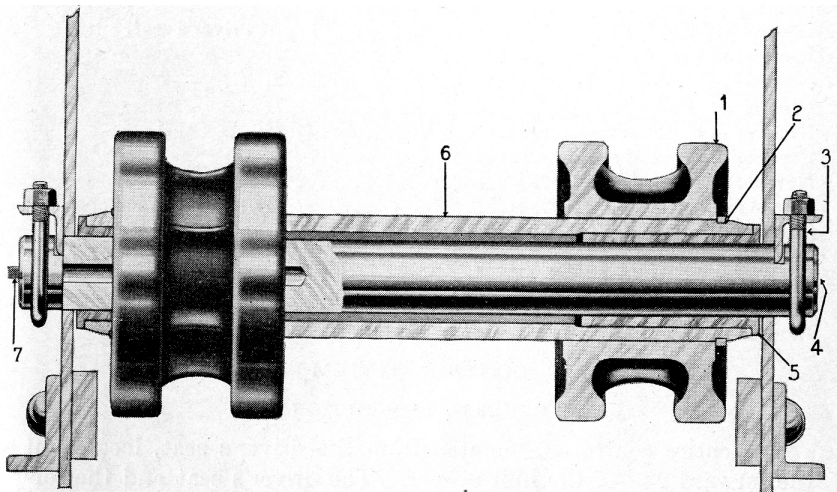
Remove the roller from the machine in accordance with directions given alone. Apply pressure on the road track roller spring plate, compressing the spring and removing the tension on the road track rollers. Separate the nuts of the spring steel split spring on the roller pin outside the roller until it is expanded to pass over the shoulder in the road track roller pin. The road track roller and roller spring plate can then be removed and the spring taken off.

TO REMOVE ROLLER PINION

After removing the track and drive wheel, unbolt the chain casing and disconnect and remove the drive chain. Unscrew the nuts holding the shaft bearing (outer end).

Insert screws in the tap holes in the shaft bearing (outer end) and screw them in, forcing the shaft bearing out. After the shaft bearing has been removed the shaft can be drawn out and the roller pinion removed and the floating bushings taken out.

Plate No. 91



LOWER TRACK ROLLER WITHOUT SPRING

Ref. No.	Part No.	Name	Ref. No.	Part No.	Name
1	M-1332	Roller.	5	M-1339	Floating bushes.
2	M-1335	Spring ring.	6	M-1333	Tube.
3	M-1338	U bolt.	7	M-1410	Tap plug in roller pin for oil.
4	M-1337	Roller pin.			

Chapter VIII

CONTROL SYSTEM

Outline specifications

Control location	At driver's seat.
Number of brake pedals	1.
Number of shifter levers	2.
Number of reverse levers	1.
Number of clutch levers	1.
Number of forward speeds	2.
Number of reverse speeds	2.
Method of steer	By relative track speeds.
Spark control	Hand lever.
Throttle control	Hand lever.
Total high-speed ratio	32.545 to 1.
Total low-speed ratio	126.64 to 1.
Reverse ratio	Same as forward.

CONTROL SYSTEM

GENERAL DESCRIPTION

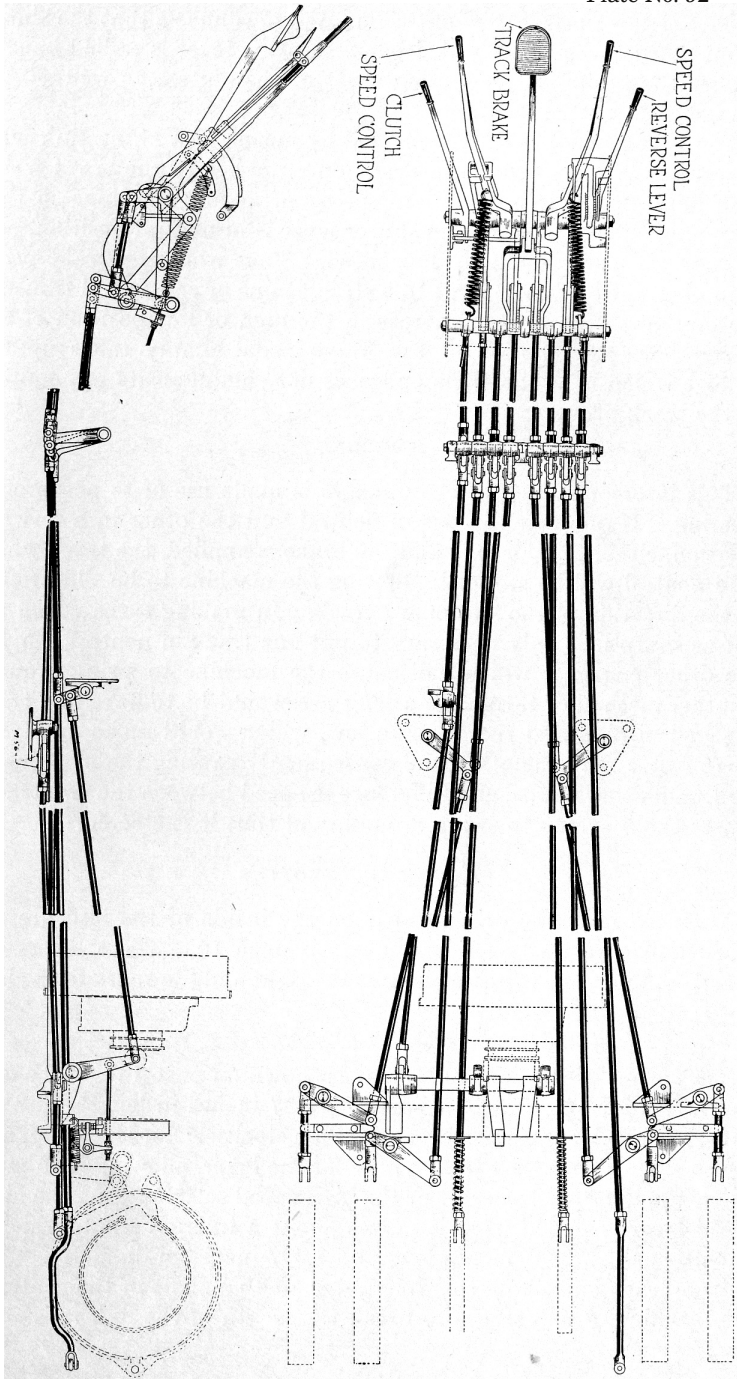
The entire control is operated from the driver's seat, located at the forward end of the hull proper. The driver's seat and the forward control unit, including the necessary levers, shafts, and linkage, are made up as a single assembly. The driver has a pedal and four levers to operate, exclusive of the spark and throttle levers. The pedal is the brake. The inner left lever is a speed control for the left track and the inner right lever is the speed control for the right track. The outer left lever is the clutch and the outer right lever the reverse.

The change-speed levers are similar for the left and right tracks. The lever engages in two quadrant jaws, the outside taking care of high speed and the inside taking care of low speed. When the lever is engaged with the high or low speed jaw of the quadrant and pulled back, the speed is engaged. Directions for starting and stopping are given on page 14, together with driving instructions. A description of the method of operation of the control system is taken up in this chapter.

SPEED CONTROLS

High or low speed controls and reverse are accomplished by moving the speed control and reverse levers to the proper position, which in turn moves a series of links which apply the high-speed or low-speed brakes in the epicyclic gear, or, in the case of the reverse, mesh the driving clutch on the epicyclic shaft so that the proper bevel ring gear is driven.

LAYOUT OF THE CONTROL SYSTEM SHOWING LINKAGE AND LEVERS



The clutch lever operates the compound clutch by means of linkage which extends to the throw-out-collar, and so connects that the clutch is out when the lever is pulled back. As the lever is pulled back it applies the brake on the cardan shaft of the clutch, as well as disengaging the clutch itself.

The brake pedal is interconnected by means of a swing link with the shifter levers so that the brake only applies to the track which is in neutral. This is accomplished by arranging the links so that they are end to end when the shifter lever is in neutral position, and at an angle when in high or low speed. Thus when the brake pedal is applied, the links which are in a straight line or end to end transmit the motion to the brake, whereas in the high or low position of the shifter lever the motion of the brake pedal simply takes up the linkage which is at the angle and does not communicate any motion to the track brake.

STEERING

This interconnection of the linkages is made use of to accomplish steering. If one track is put in neutral and the other in a forward position, either high or low, and the brake is applied, the track which is in neutral will be stopped, allowing the machine to be slid around in the direction of the stationary track. In making turns which are not as sharp it is only necessary to put one track in neutral, leaving the other engaged, which will cause the machine to swing around. Another method of turning in a long arc would be to have one track engaged in high and the other in low, which would make one track move faster than the other and consequently causing the machine to turn. In other words, any difference in speed between the two tracks causes the machine to swing around, and thus it is steered.

SPARK AND THROTTLE

Mounted near the driver's seat on the inside of the hull are the spark and throttle levers as shown on page 16. These levers are exactly similar, the throttle being to the right and the spark to the left.

CONTROL LINKAGE

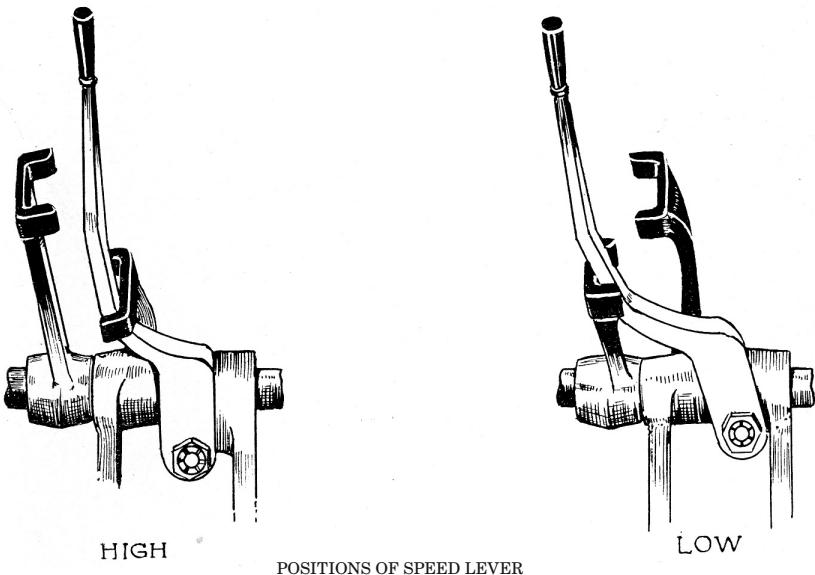
The layout of the control linkage is shown on page 147. The control levers for operating the speeds are 37 inches in length, and are of carbon steel, engaging with the carbon steel selector levers. These are so shaped that they form a gate for the lever, allowing it to move either the low-speed selector or the high-speed selector. As the high-speed lever is pulled back it moves about a fulcrum shaft, which is of carbon steel, $24\frac{3}{4}$ inches long and 1.497 inches in diameter. The linkage leads back from this fulcrum shaft to a shaft in the center of the machine, which serves to break the length of the link rods. By

means of clevis connections and toggle joints the control is passed along backward to the high and low speed brake toggles, which operate the control brakes on the high and low speed drums of the epicyclic gear.

BRAKE CONNECTIONS

The track brakes are on each side of the epicyclic, just inside the chain sprocket, on the epicyclic shaft. As briefly explained, the linkage of the brake is so arranged that only the brake on the side in neutral is applied. The linkage is so interconnected that when the shifter lever is moved to neutral, the brake bands just clear the drum. Application of pressure on the pedal, therefore, results in immediate

Plate No. 93



pressure on the drum. The brake pedal is a malleable-iron casting $35\frac{1}{4}$ inches long over all, the pedal being 8 inches long by 5 inches wide, I-section. The connections of this pedal, which is a bell crank, connected with a series of toggles and links, are clearly illustrated on page 147. The brake rods are held back in three sections as in the case of the other control rods, so as to break their length, and at the same time secure the necessary reduction.

THE CLUTCH LEVER

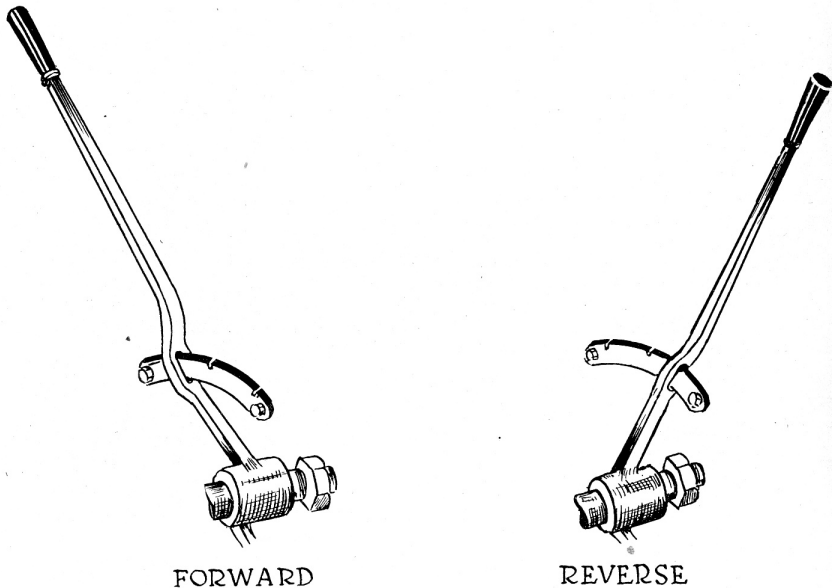
The clutch lever is a carbon steel lever in the form of a bell crank. The hand end is $28\frac{1}{2}$ inches long from the center of the fulcrum to the end. The bell-crank end is 5 inches long. This connects with the toggle joint, mounted on the shaft, directly behind the lever

fulcrum shaft, and the rods extent backward to the main clutch operating the clutch throw-out lever, which removes the cone from engagement with the flywheel and also the driving box on the clutch shaft.

REVERSE LEVERS

The reverse lever is a carbon steel piece of bell-crank form, being 27.968 inches long on the hand side of the fulcrum and 6 inches on the bell crank. It engages with the reverse rod, which is 52 inches long, and made of extra strong $\frac{3}{4}$ -inch wrought-iron pipe. Similar to the other control rods, this is in three sections, front, center, and rear, carrying the motion of the reverse lever back to the rear control

Plate No. 94



POSITIONS OF REVERSING LEVER

assembly, where the necessary motion is imparted to operate the shifter fork for the clutch in the epicyclic gear. The linkage is outlined clearly on page 147. The necessary motion for shifting the fork is transverse, and thus secured by bell-crank levers from the reverse rods.

TO ADJUST THE CARDAN BRAKE

The cardan brake is operated by means of the clutch stop rod. This is coupled at one end to the left clutch lever, which is threaded and has two nuts between which is the clutch-stop bell crank. Through the opposite end of the left clutch lever passes the clutch-stop eye-rod bolt with the end of the clutch band.